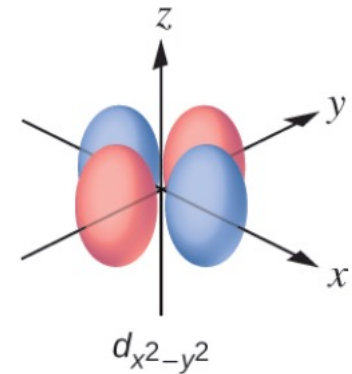
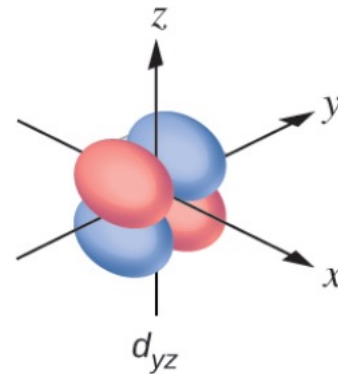
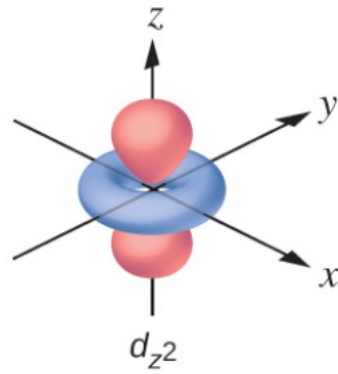
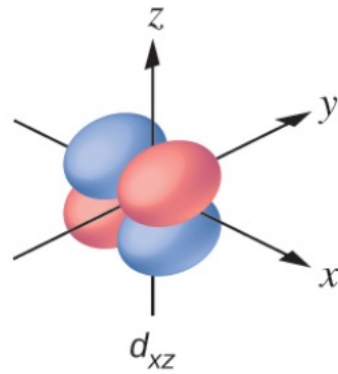
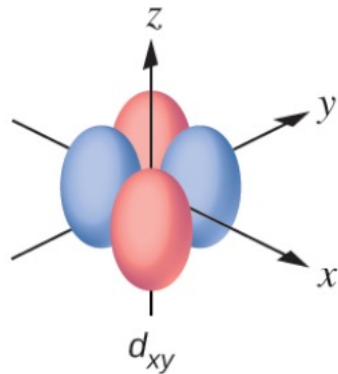
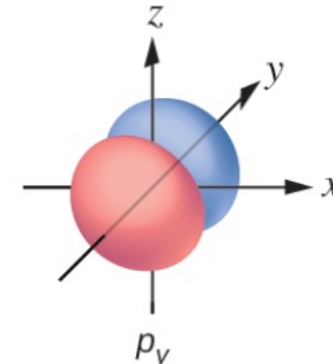
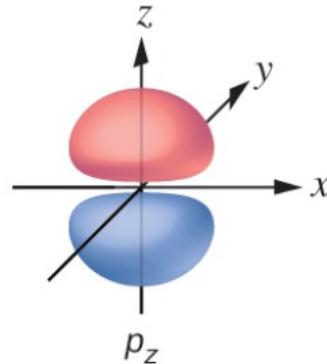
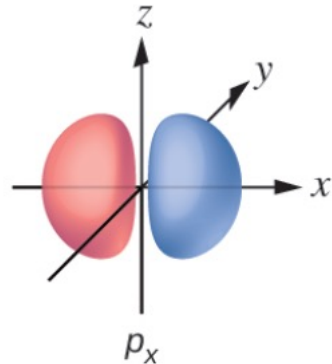
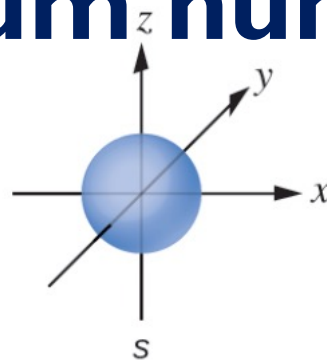
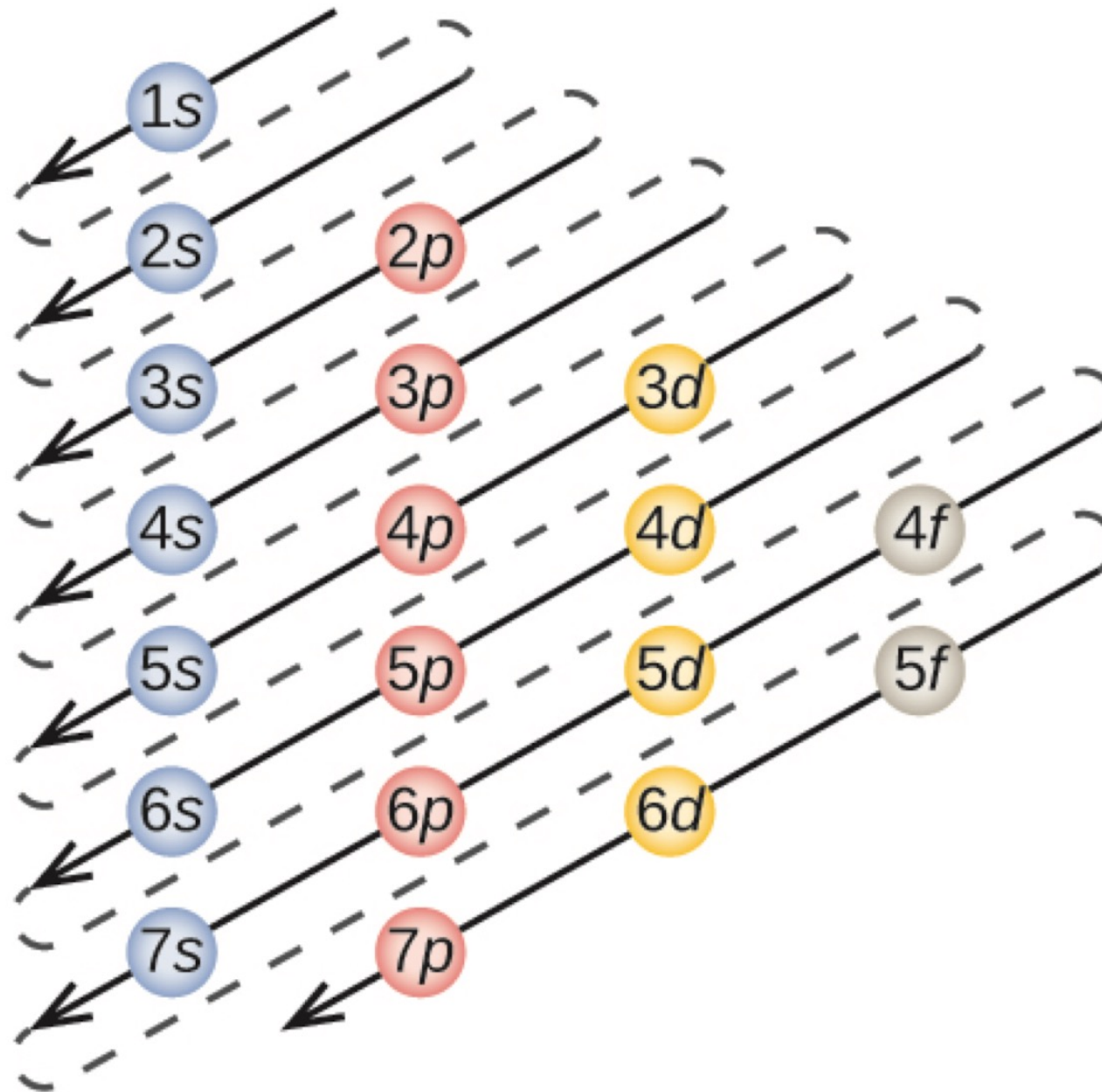


A set of quantum numbers define an orbital

Angular parts:



Orbital Energies & Aufbau Principle



Atom and ion electronic structure

How to put electrons in orbitals??

- ▶ Fill the lowest energy orbitals first

Energy increases with the sum $n + l$:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, ...

- ▶ The set of quantum numbers for an electron is unique, i.e., two electrons cannot have the same four quantum numbers.

Pauli exclusion principle

- ▶ For a given n, l quantum number the most stable state maximizes $m_{s,total} = \sum_i m_{s,i}$

Hund's rule

Example: The Oxygen Atom

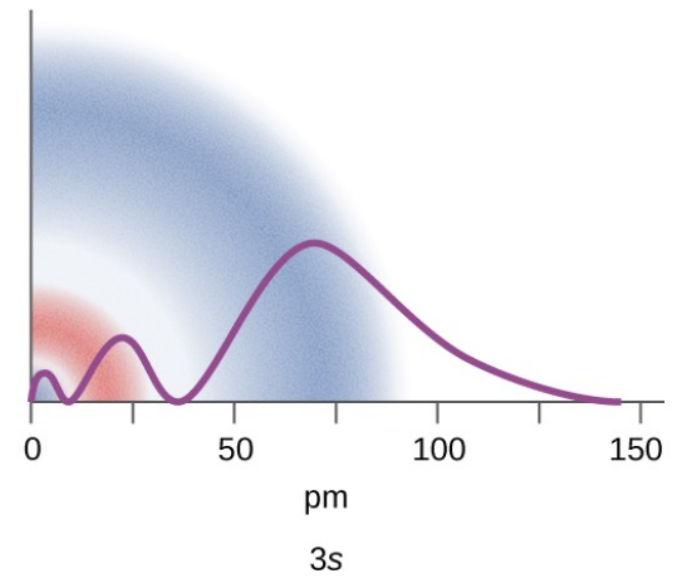
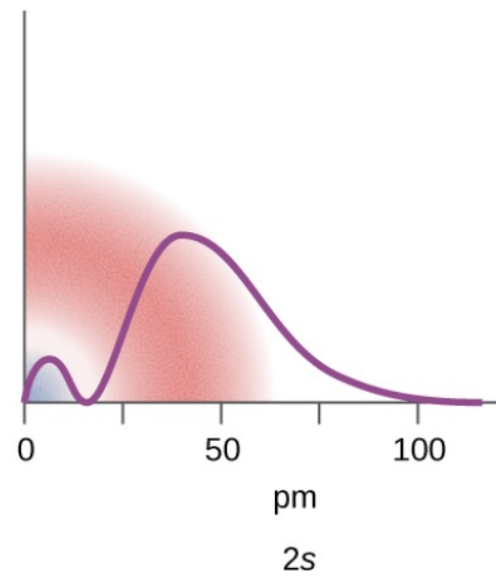
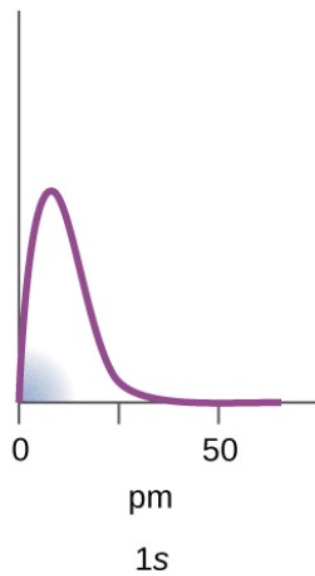
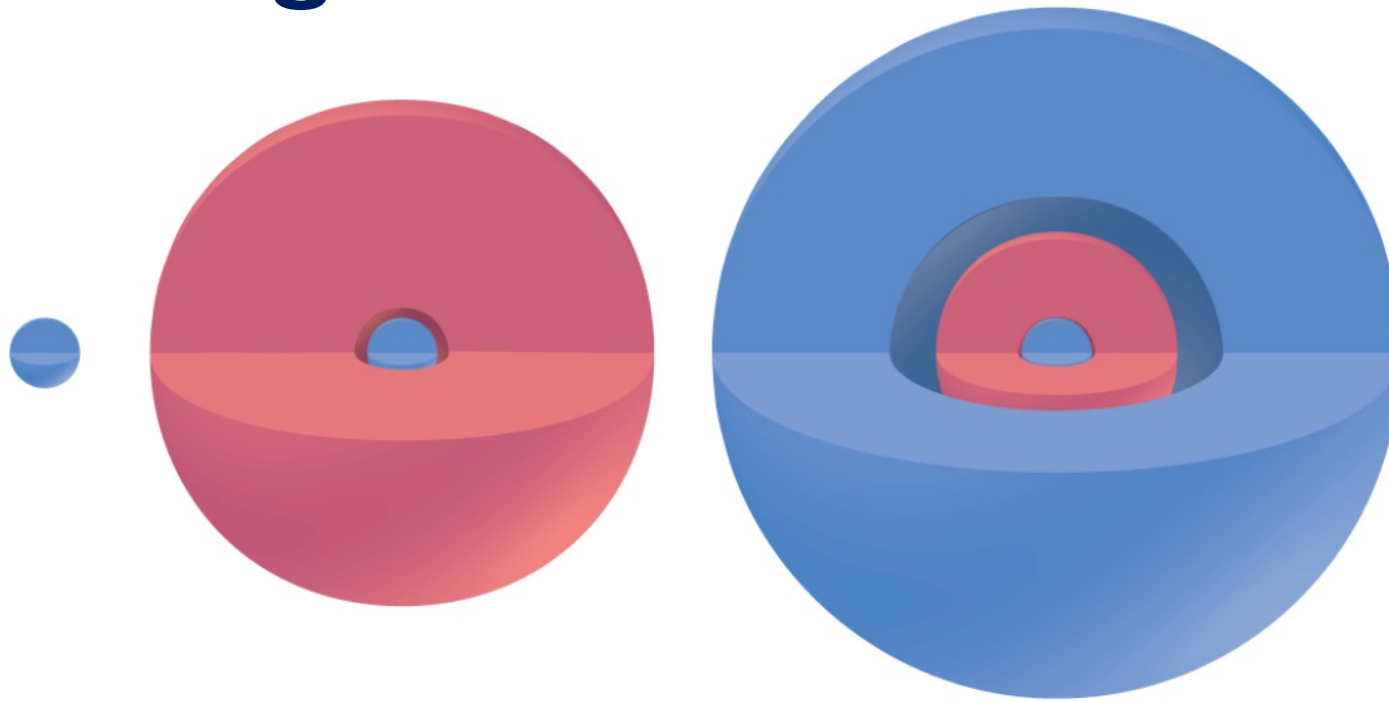
Exercise 2

What is the electronic configuration for:

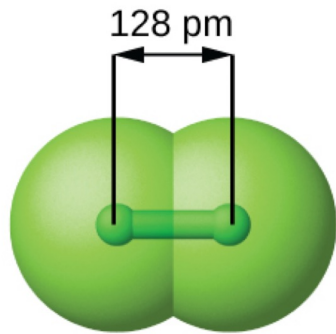
(a) Sc, (b) Se^{2-} ?

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* * 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

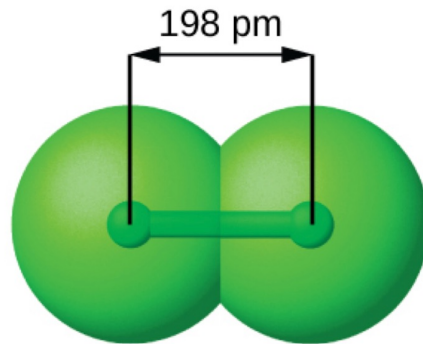
How big is an atom?



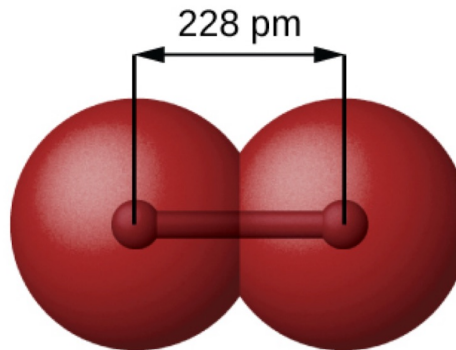
How big is an atom?



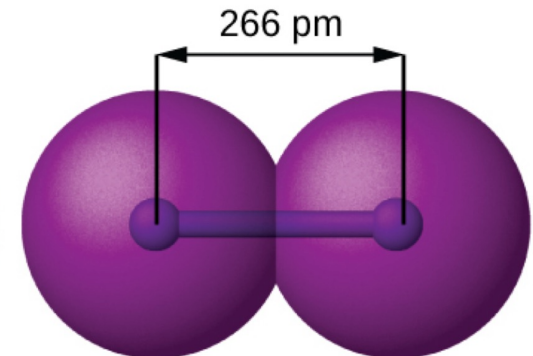
$$\text{F radius} = \frac{128 \text{ pm}}{2} = 64 \text{ pm}$$



$$\text{Cl radius} = \frac{198 \text{ pm}}{2} = 99 \text{ pm}$$



$$\text{Br radius} = \frac{228 \text{ pm}}{2} = 114 \text{ pm}$$



$$\text{I radius} = \frac{266 \text{ pm}}{2} = 133 \text{ pm}$$

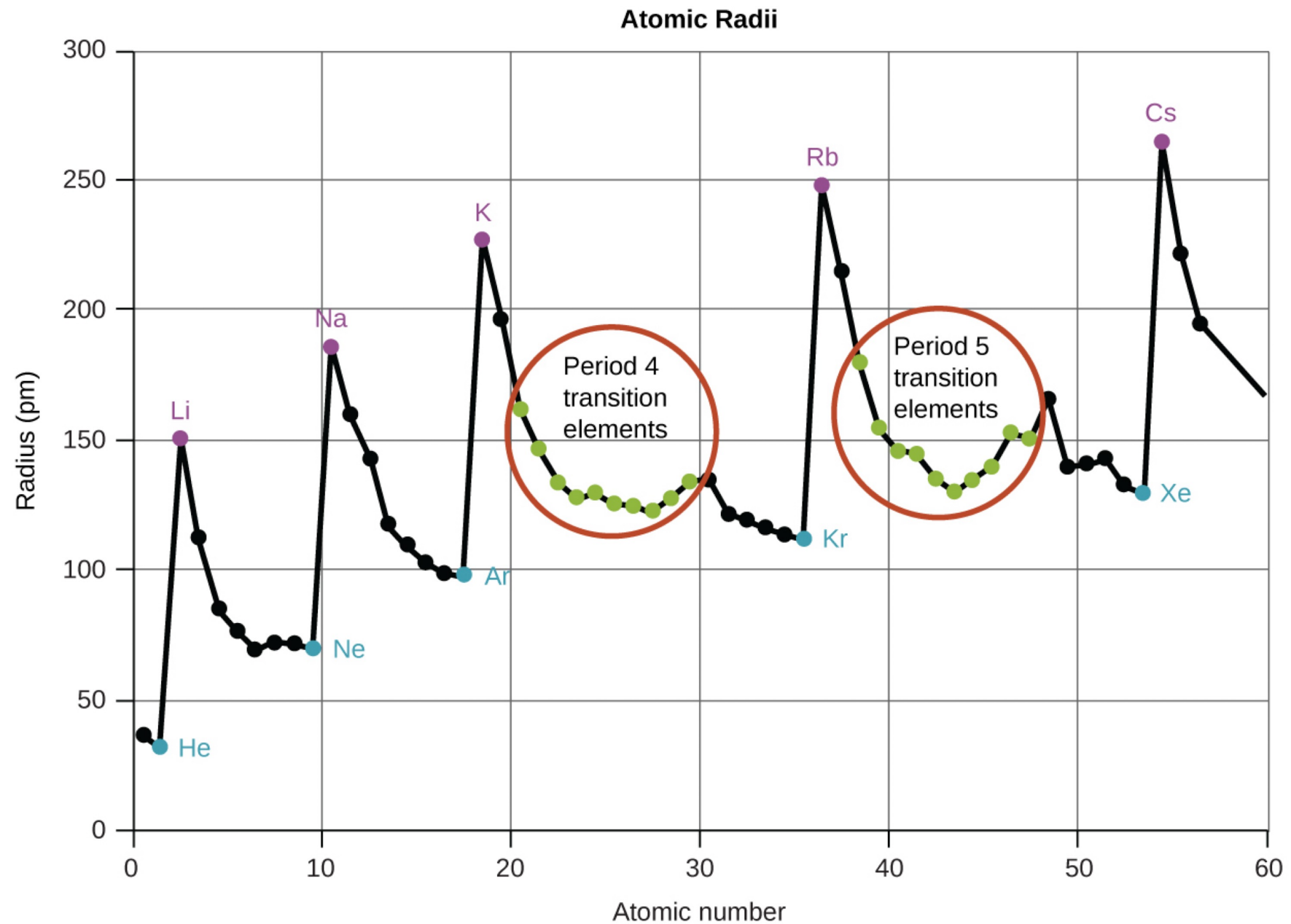
Effective nuclear charge (Z_{eff})

Across a period:

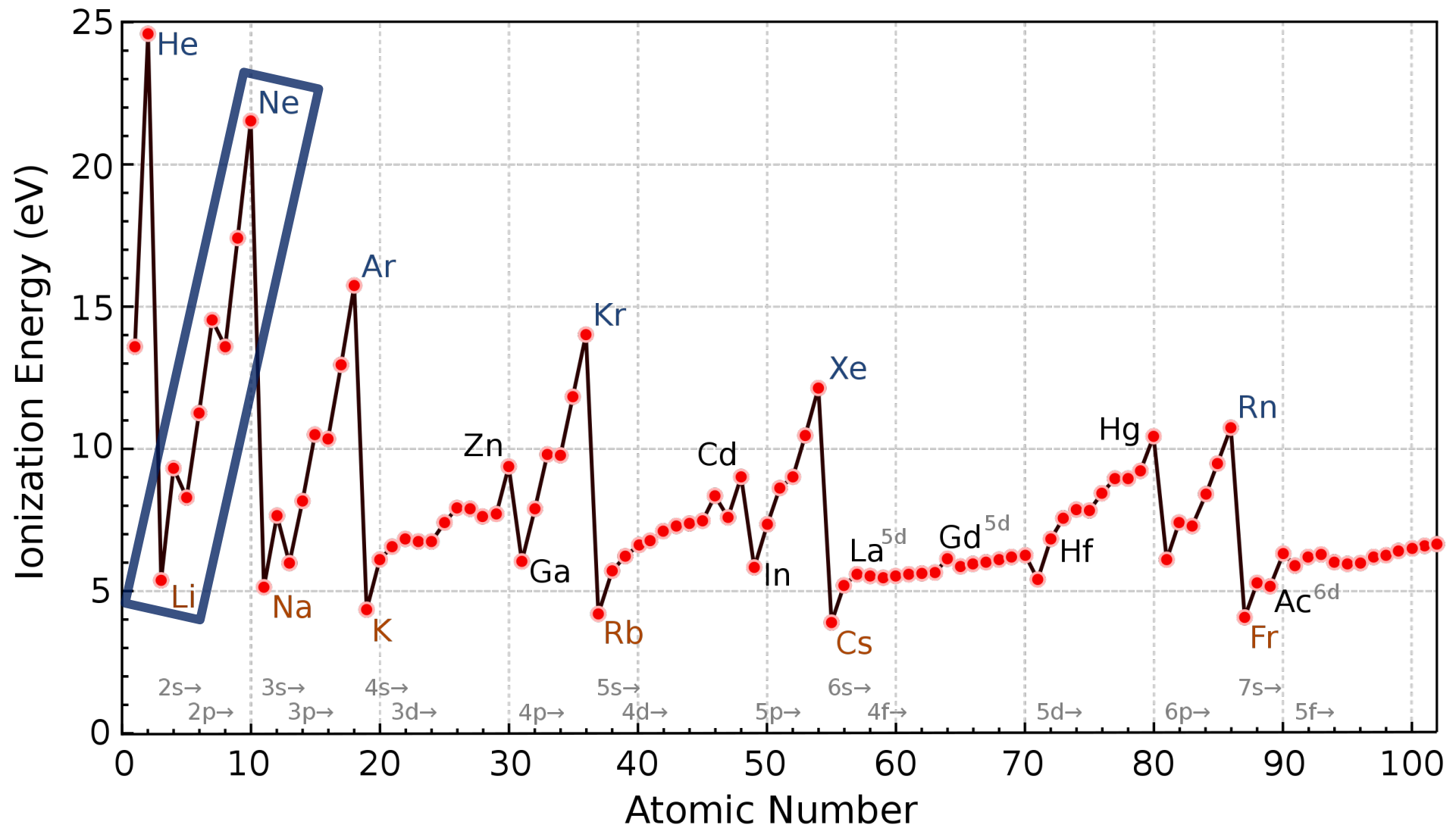
	Li	Be	B	C	N	O	F	Ne
Z	3	4	5	6	7	8	9	10
Z_{eff}	1.3	1.9	2.4	3.1	3.8	4.5	5.1	5.8

Atomic radius *decreases* along the period(!)

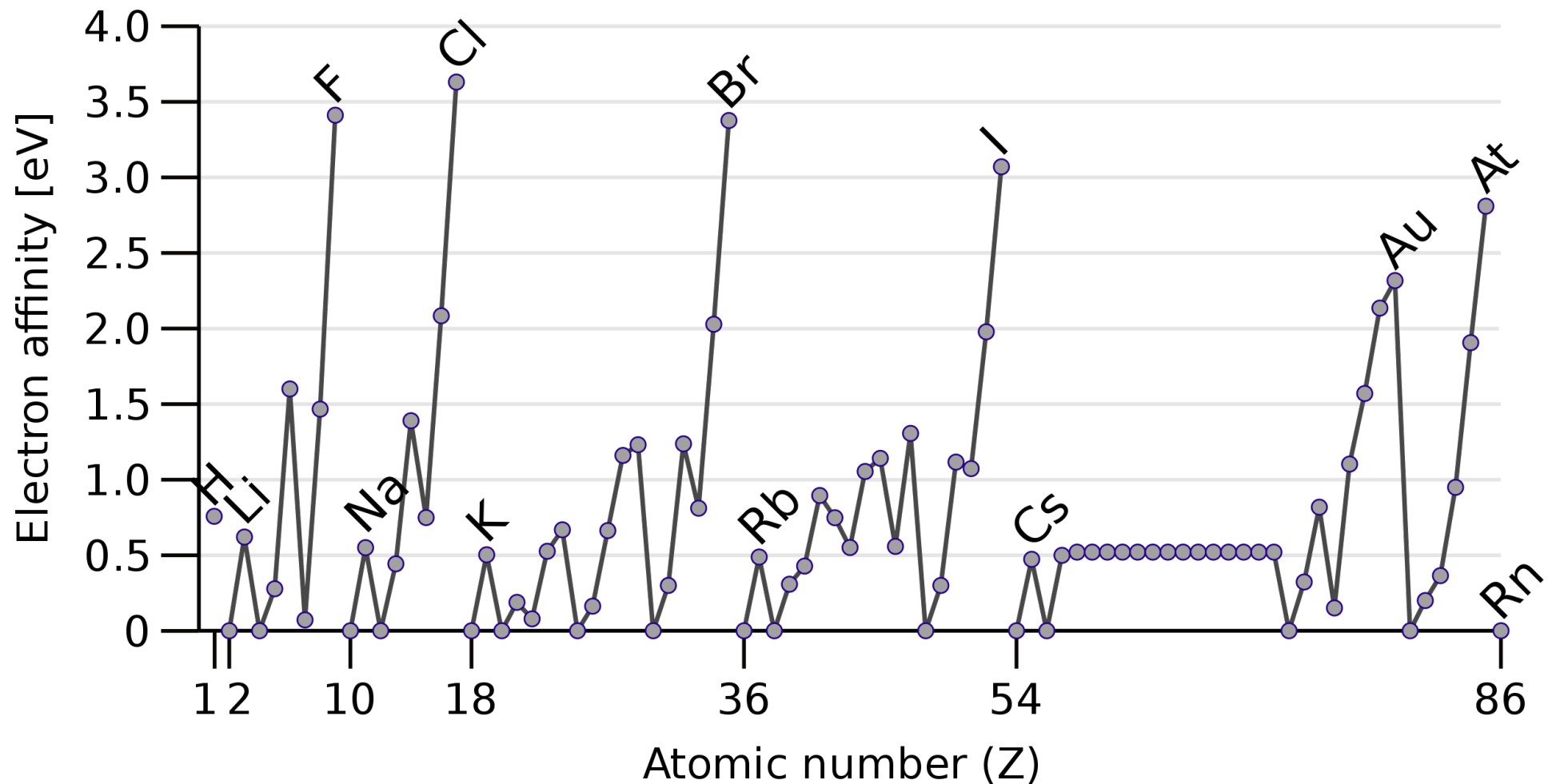
Atomic radius – along periods



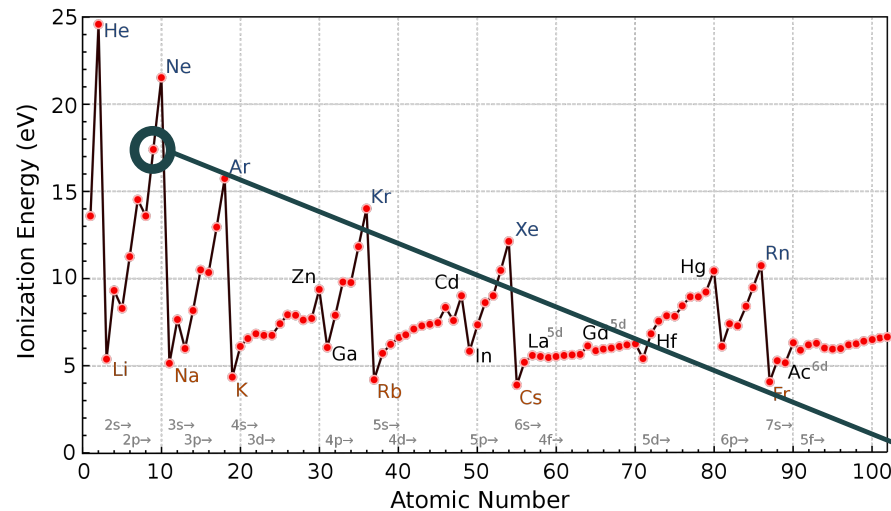
Ionization energy



Electron affinity

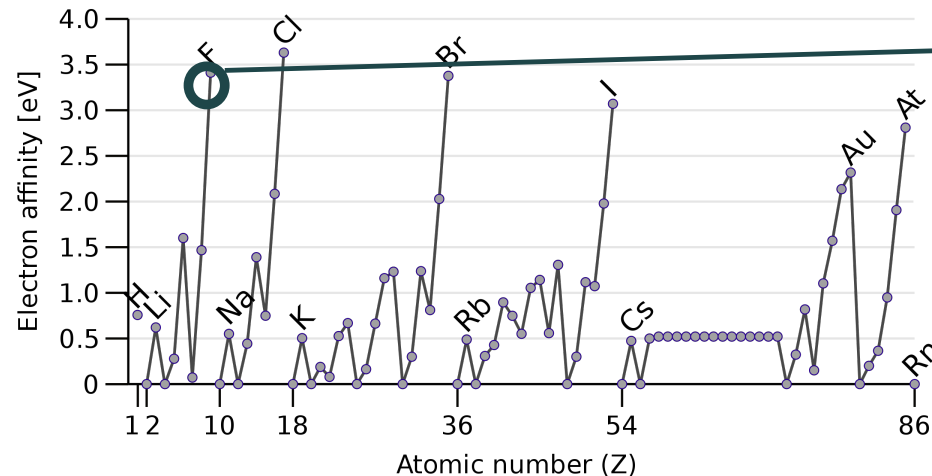


Electronegativity



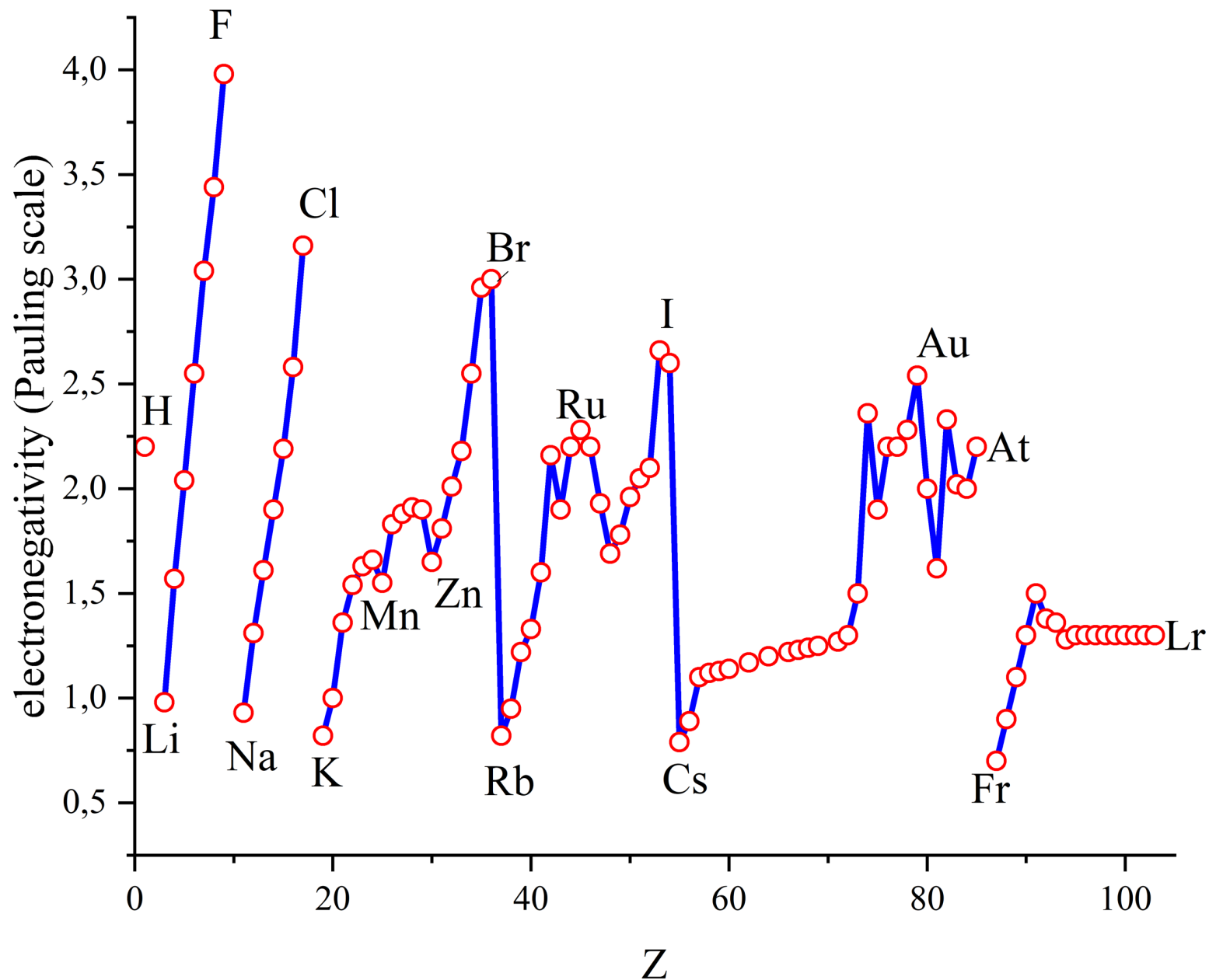
Mulliken's (simple) definition

$$\chi = (IE + EA)/2$$

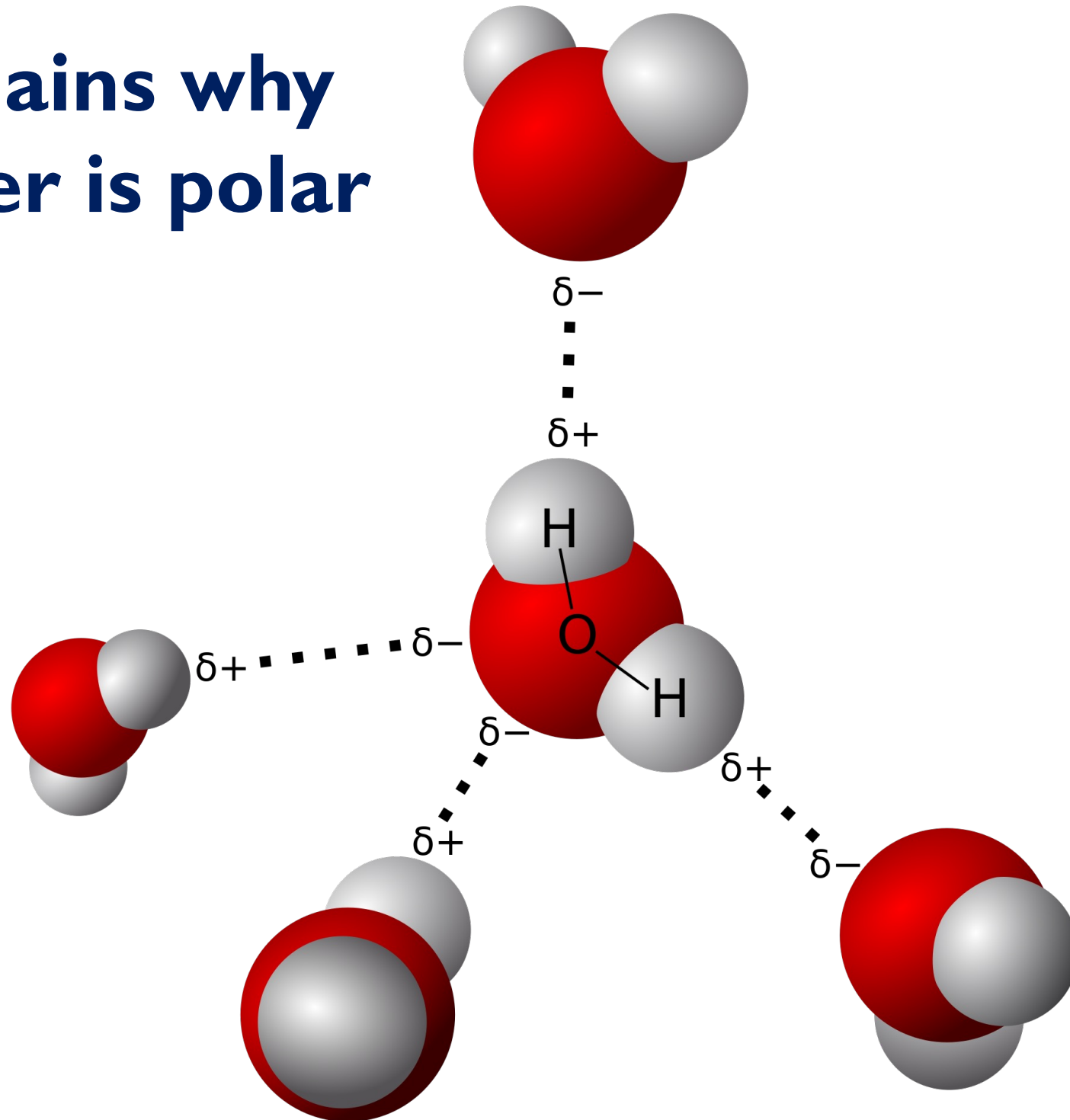


Fluorine is the most electronegative atom

Electronegativity



Explains why water is polar



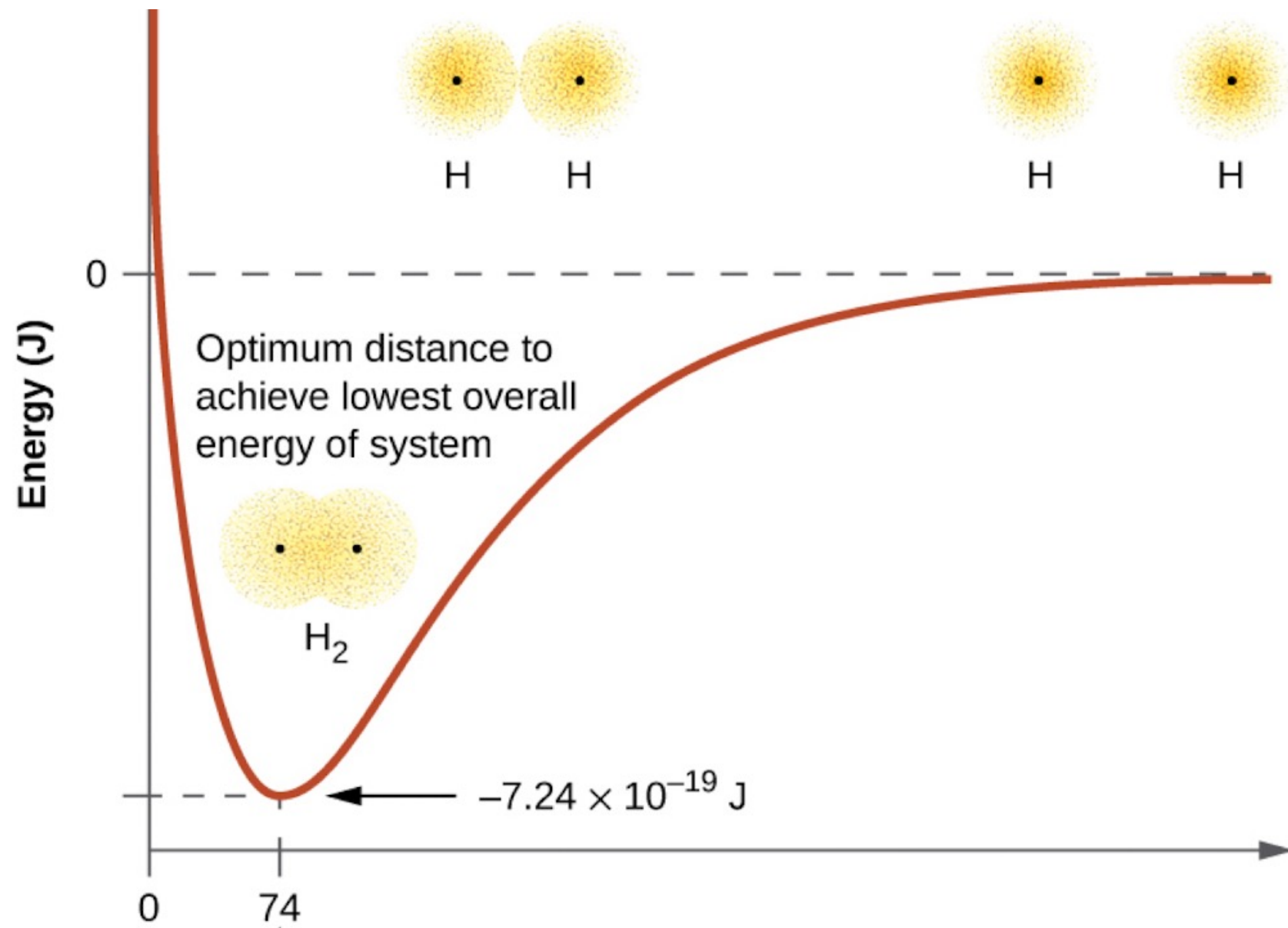
Chemical bonds, geometries and hybridization

2602I Chemistry (Polytechnical Foundation)

Professor Kasper S. Pedersen

2026

The simplest molecule: H₂

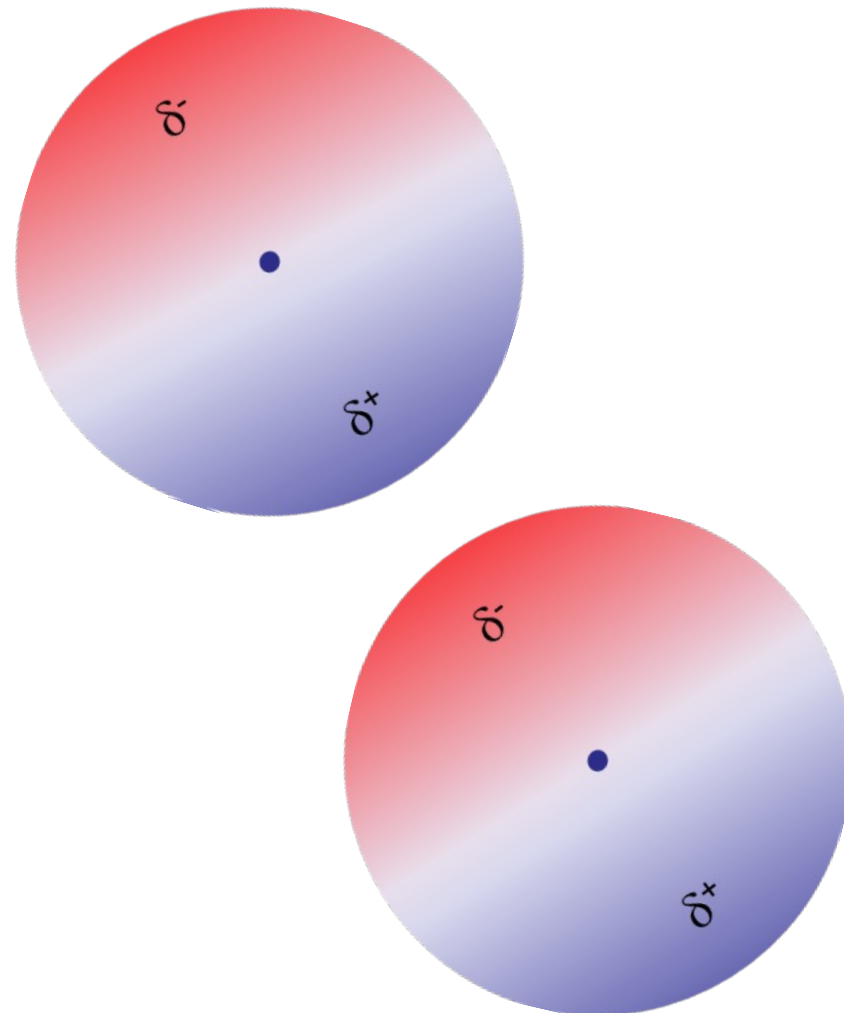


Van der Waals interactions

Weakest chemical bonds: between permanent dipole, permanent dipoles and induced dipole, induced dipoles.

Bond energy < 10 kJ/mol

E.g.: noble gas liquids and crystals (He, Ne, Ar, ...), interactions between molecules



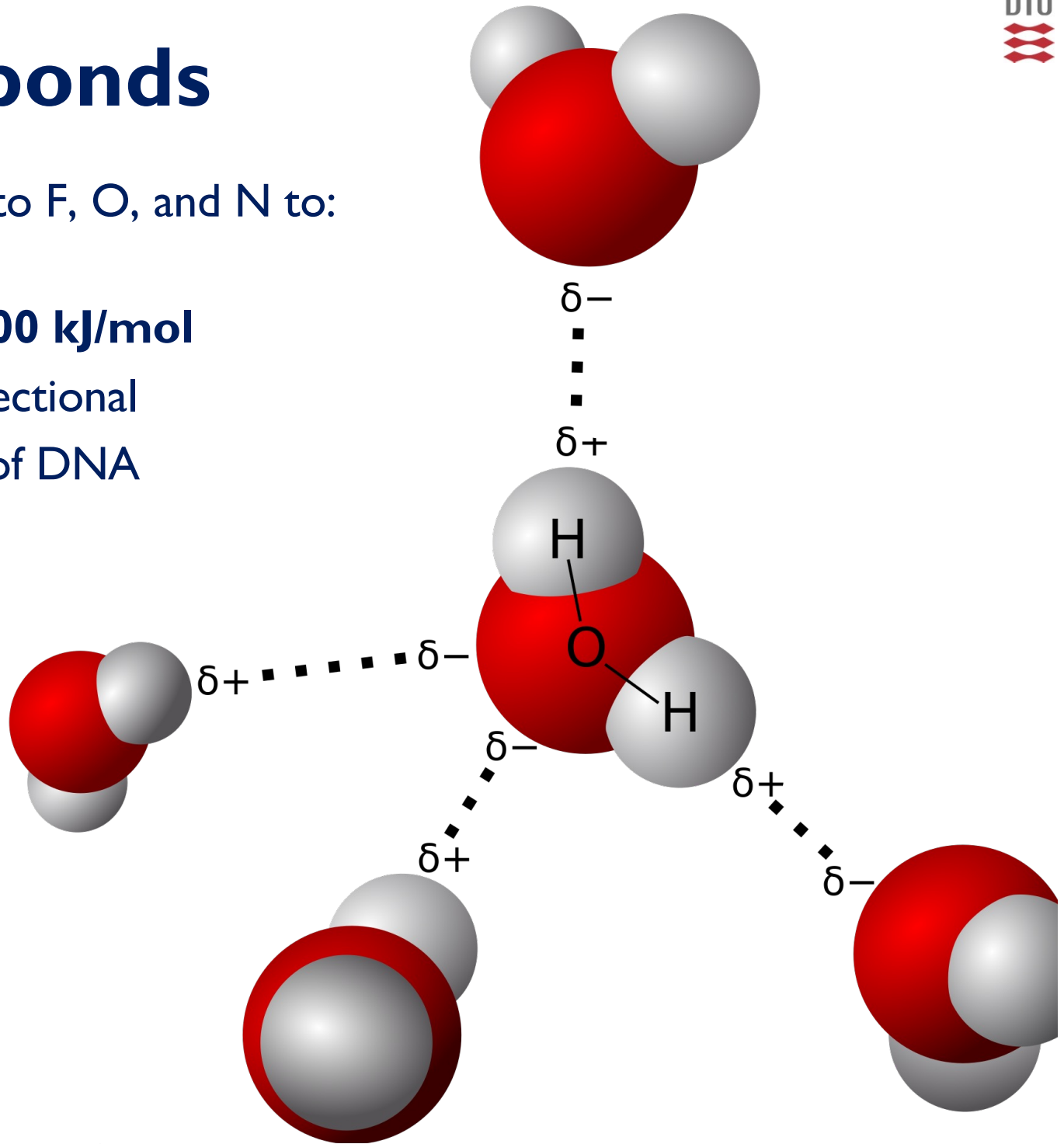
Hydrogen bonds

Only from H bonded to F, O, and N to:
F, O, and N.

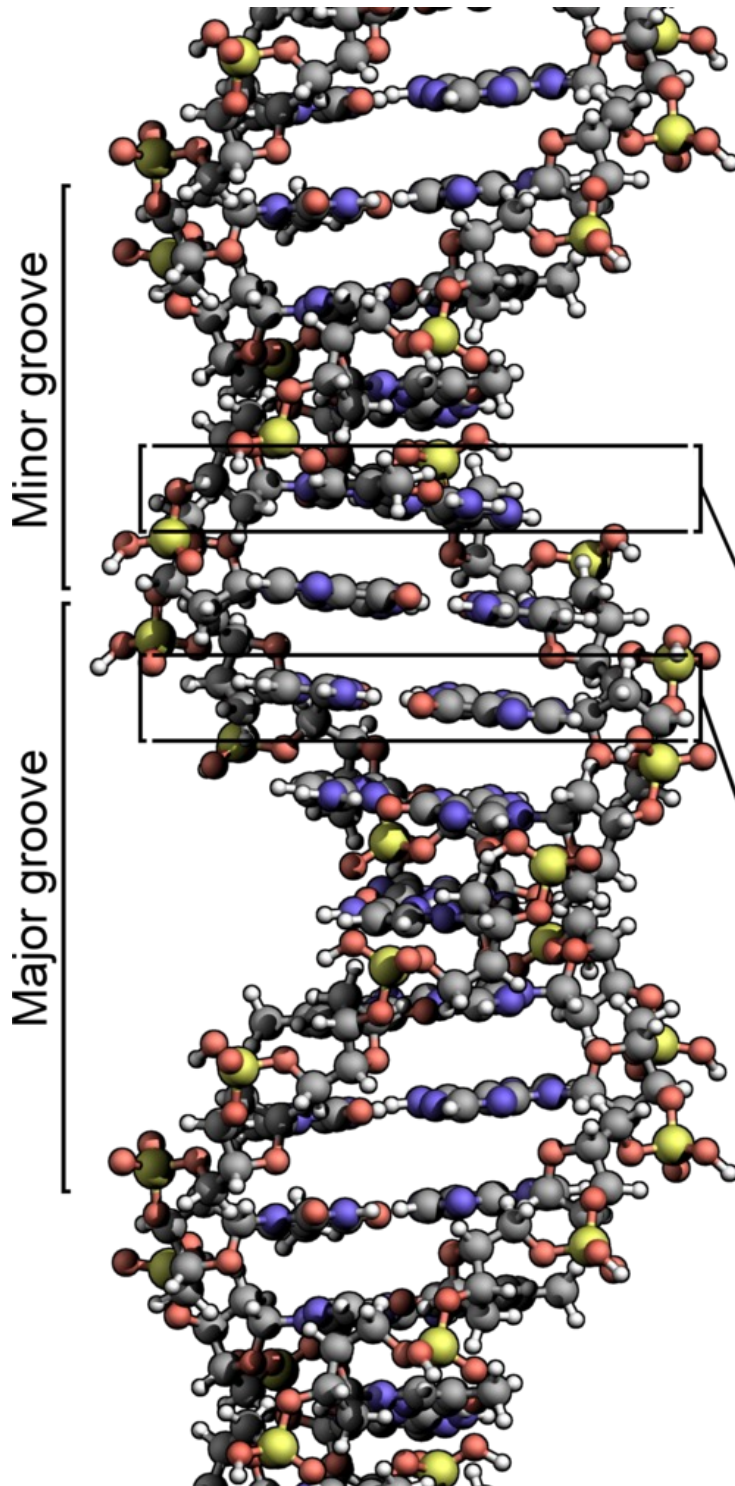
Bond energy ~10-100 kJ/mol

Very localized and directional

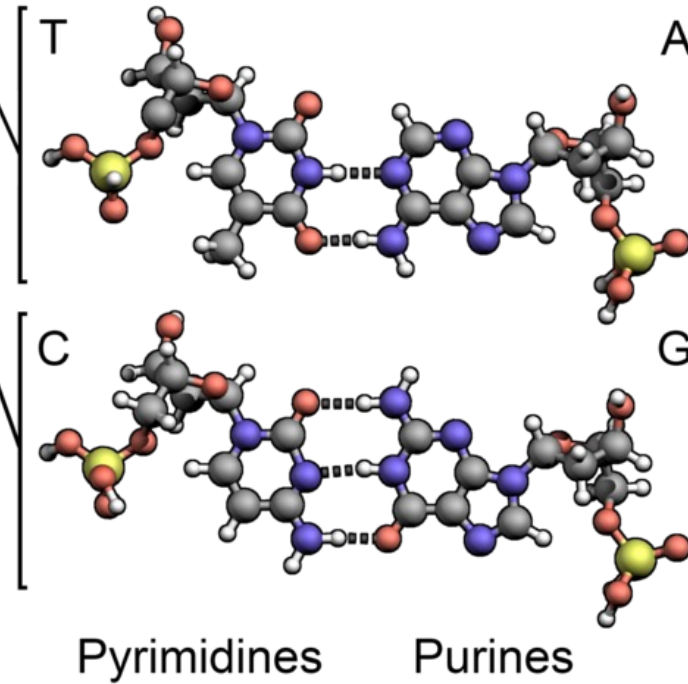
E.g.: water, structure of DNA



Hydrogen bonds



- Hydrogen
- Oxygen
- Nitrogen
- Carbon
- Phosphorus

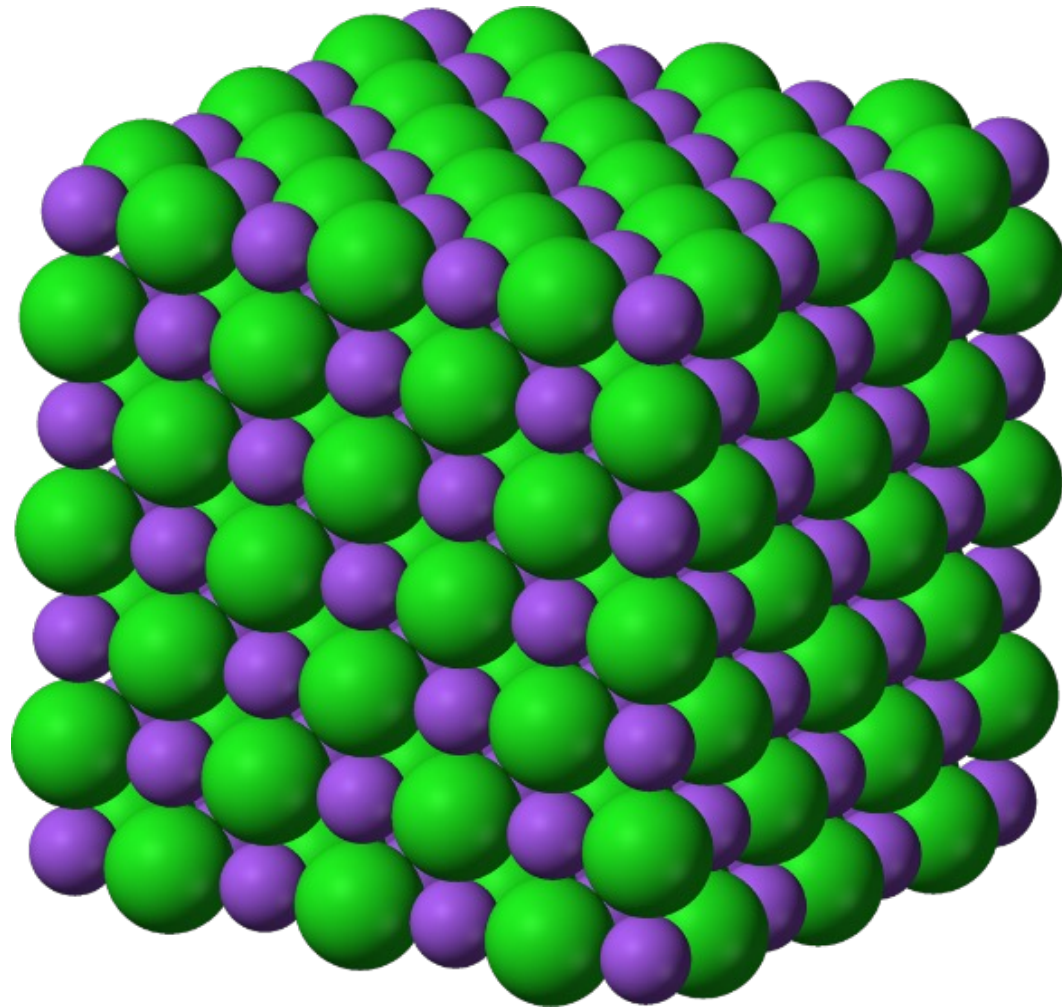


Ionic bonds

Electrons are localized – thus low electric conductivity.

Bond energy **~400-4000 kJ/mol**; electrostatic interaction (Coulomb's Law)

E.g. MgO, NaCl,...



Ionic radius

Sizes of atoms and their ions in pm

Group 1		Group 2		Group 13		Group 16		Group 17	
Li^+	Li	Be^{2+}	Be	B^{3+}	B	O	O^{2-}	F	F^-
90	134	59	90	41	82	73	126	71	119
Na^+	Na	Mg^{2+}	Mg	Al^{3+}	Al	S	S^{2-}	Cl	Cl^-
116	154	86	130	68	118	102	170	99	167
K^+	K	Ca^{2+}	Ca	Ga^{3+}	Ga	Se	Se^{2-}	Br	Br^-
152	196	114	174	76	126	116	184	114	182
Rb^+	Rb	Sr^{2+}	Sr	In^{3+}	In	Te	Te^{2-}	I	I^-
166	211	132	192	94	144	135	207	133	206

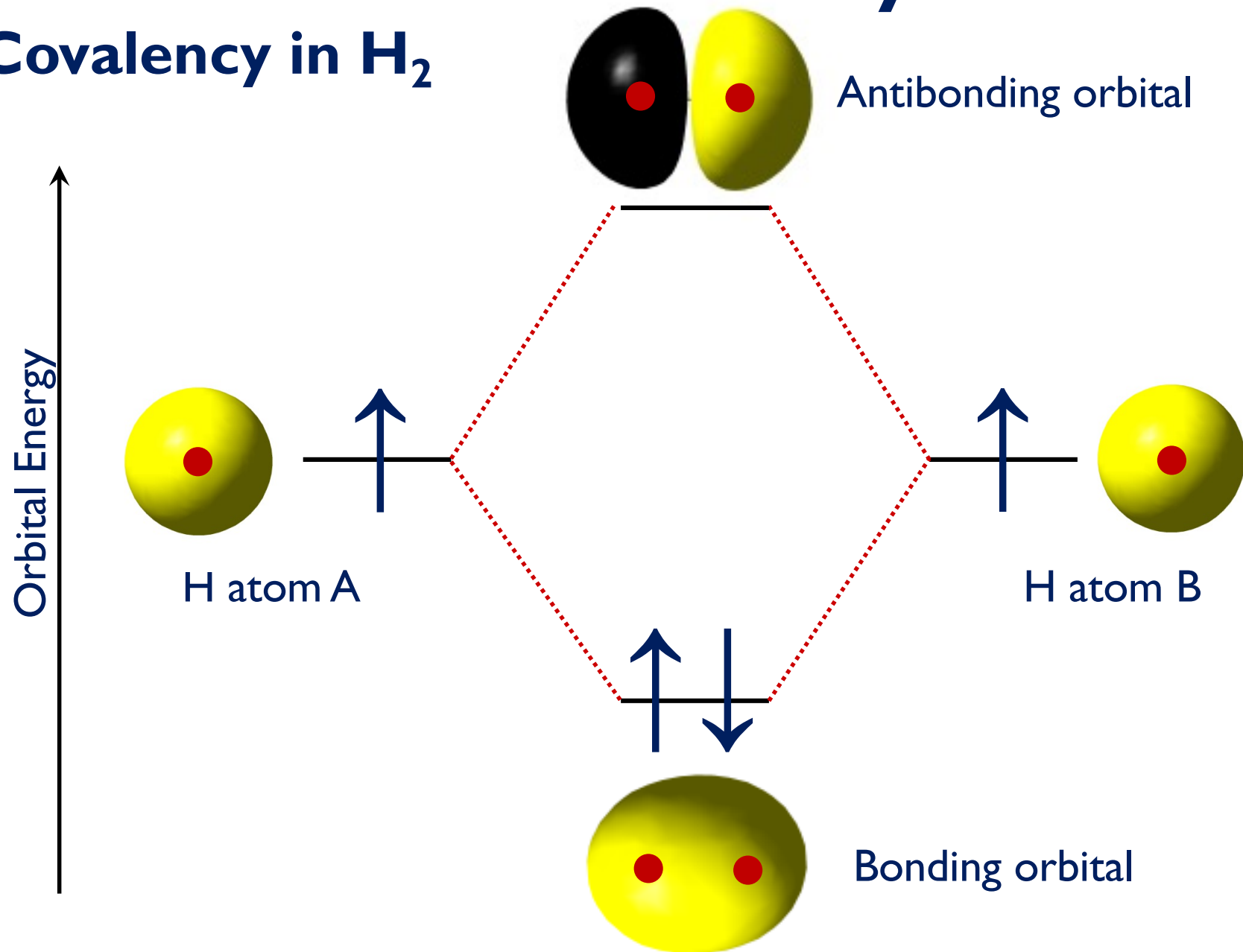
Why NaCl does not exist

Born-Haber cycle (Hess's law) for NaCl (energies in kJ mol⁻¹)



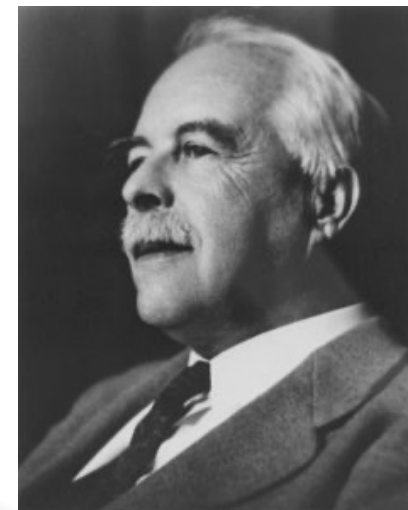
Molecular orbital theory

Covalency in H_2

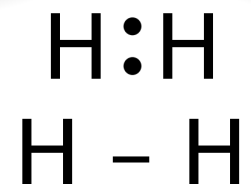
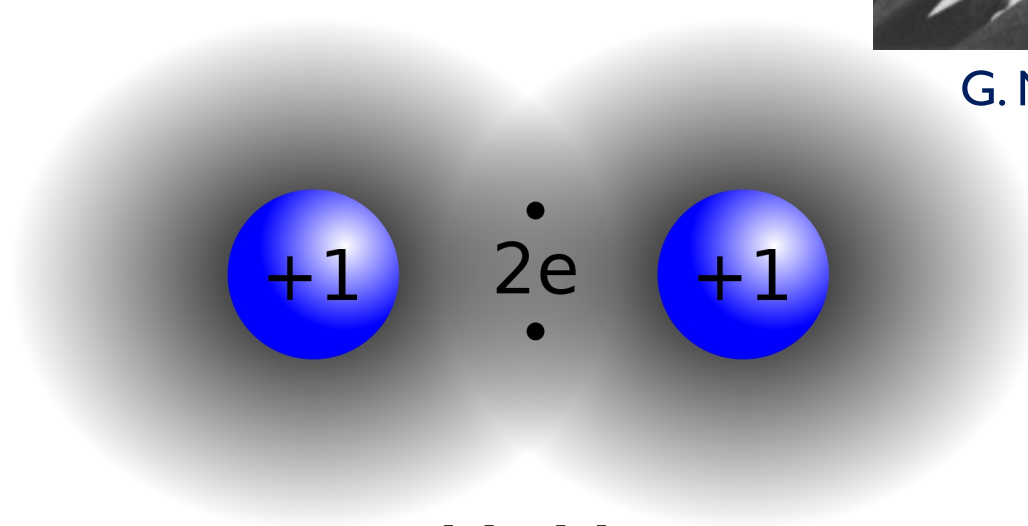
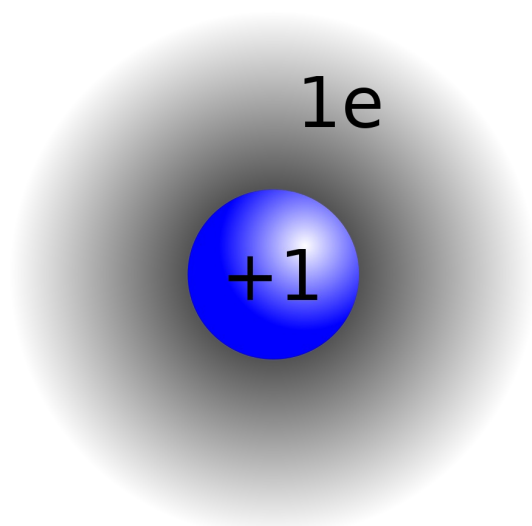


Lewis structures

Lewis structures are diagrams that show the bonding between atoms of a molecule, as well as the lone-pairs that may exist in the molecule



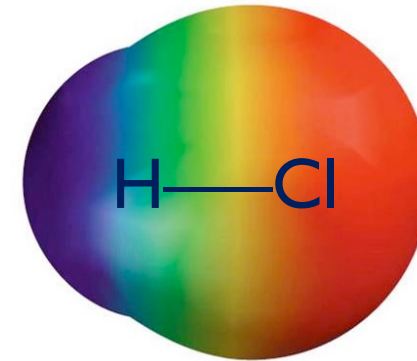
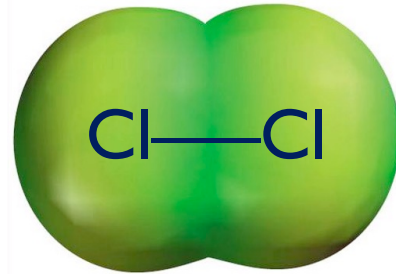
G. N. Lewis



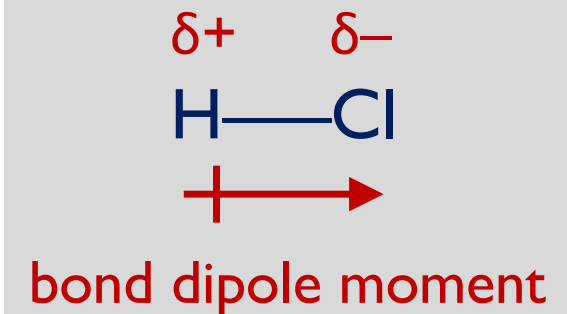
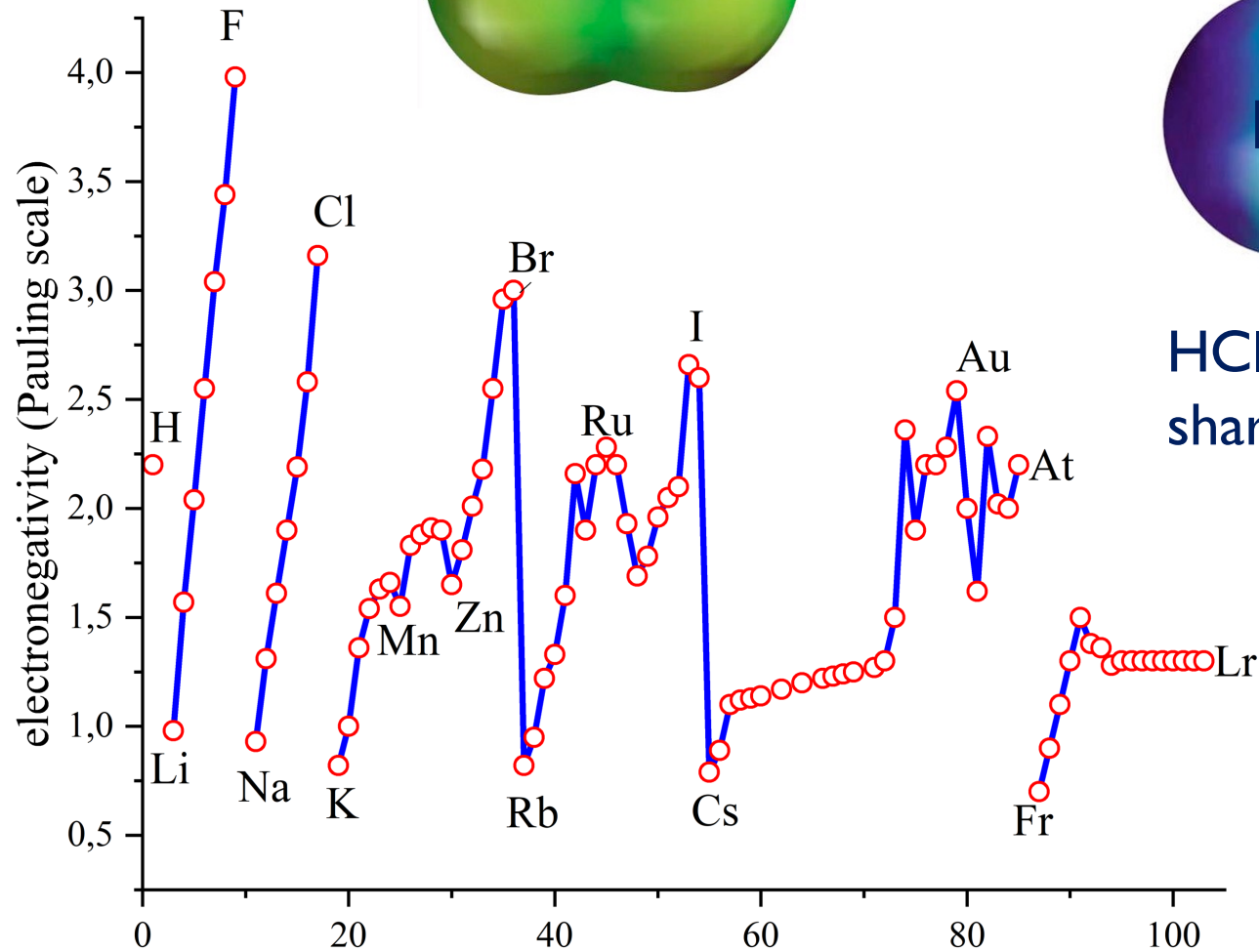
Exercise: How would the situation be for F₂?

Electronegativity & Polarity

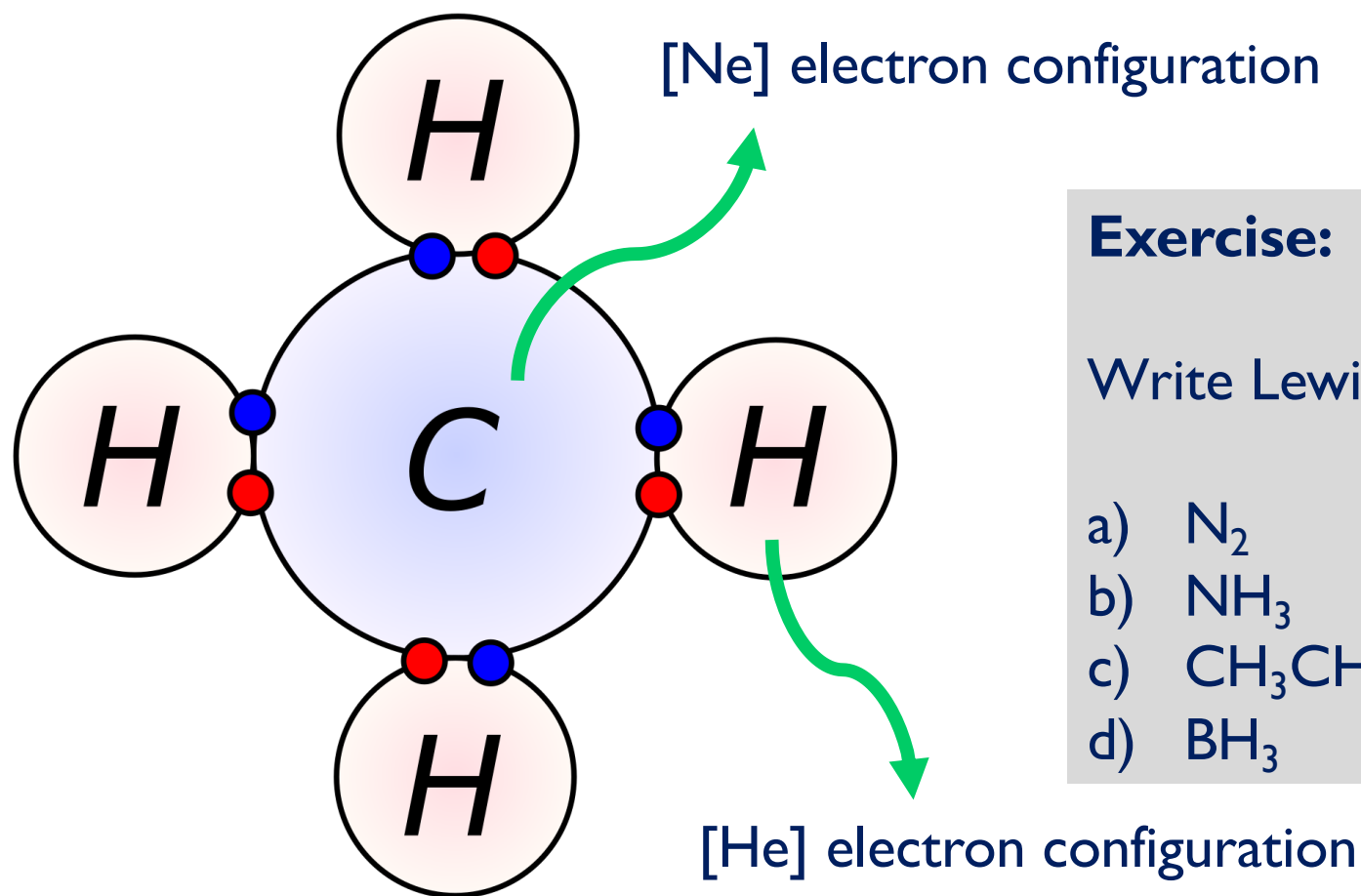
Cl_2 , electrons equally shared: *non-polar*



HCl , electrons *not* equally shared: *polar*



Lewis structure of methane, CH_4



- Electron from hydrogen
- Electron from carbon

Exercise:

Write Lewis structures for:

- a) N_2
- b) NH_3
- c) $\text{CH}_3\text{CH}_2\text{OH}$
- d) BH_3

Lewis structure of methane, CH₄

Strategy:

- 1) Write skeletal structure of molecule
- 2) Count the total number of valence electrons
- 3) Draw all bonds as single bonds and fill remaining electrons to complete octets
- 4) If it does not add up, use lone-pairs to generate double or triple bonds.

Exercise:

Write Lewis structures for:

- a) N₂
- b) NH₃
- c) CH₃CH₂OH
- d) BH₃

Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1	1 H																	2 He	
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne	
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar	
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
6	55 Cs	56 Ba	*	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	*	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			*	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			*	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

Covalent bond – energies & lengths

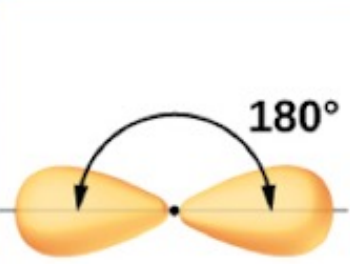
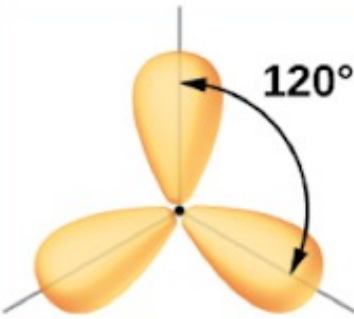
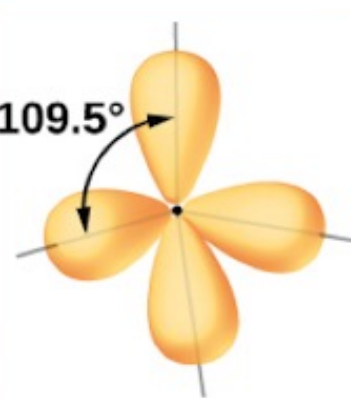
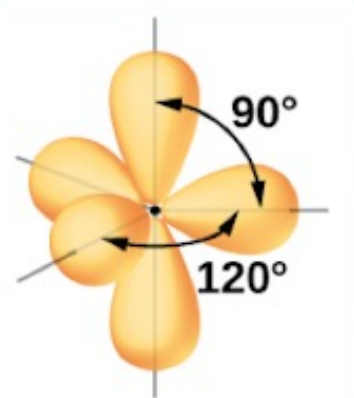
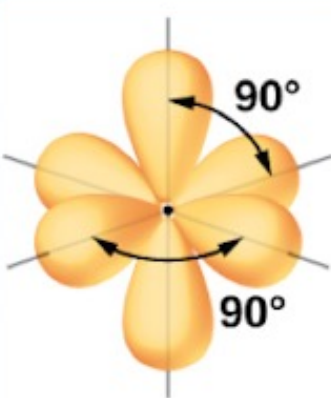
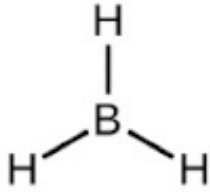
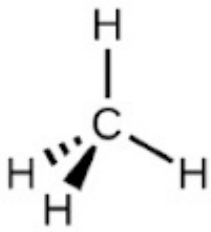
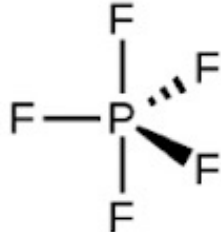
Bond Enthalpies (kJ mol ⁻¹)			
H—H	436.4	C—S	255
H—N	393	C=S	477
H—O	460	N—N	193
H—S	368	N=N	418
H—P	326	N≡N	941.4
H—F	568.2	N—O	176
H—Cl	431.9	N=O	607
H—Br	366.1	O—O	142
H—I	298.3	O=O	498.7
C—H	414	O—P	502
C—C	347	O=S	469
C=C	620	P—P	197
C≡C	812	P=P	489
C—N	276	S—S	268
C=N	615	S=S	352
C≡N	891	F—F	156.9
C—O	351	Cl—Cl	242.7
C=O	745	Br—Br	192.5
C—P	263	I—I	151.0

Bond length (Å)	
C—H	1.07
C—O	1.43
C=O	1.21
C—C	1.54
C=C	1.33
C≡C	1.20
C—N	1.43
C=N	1.38
C≡N	1.16
N—O	1.36
N=O	1.22
O—H	0.96

VSEPR Model

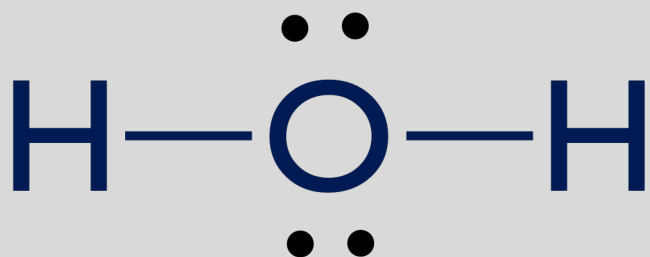
- **V**alence **S**hell **E**lectron **P**air **R**epulsion model – simple method to predict the geometry of (most) molecules.
- The idea: count the number of electron pairs around an atom. The electron pair will maximize the distance between them - i.e. each electron pair have as much room around them as possible.
- Whether the electron pair is a chemical bond or a lone pair does not matter



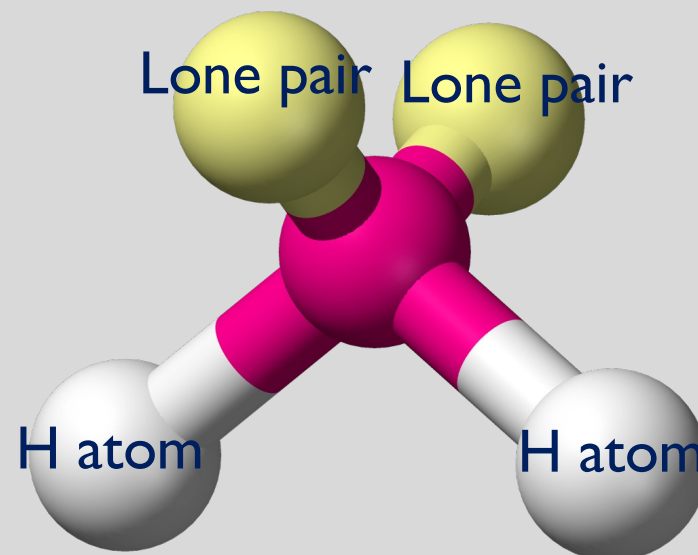
				
$\text{H}-\text{Be}-\text{H}$				
Linear; 180° angle	Trigonal planar; all angles 120°	Tetrahedral; all angles 109.5°	Trigonal bipyramidal; angles of 90° or 120°	Octahedral; all angles 90° or 180°

VSEPR Example I: H_2O

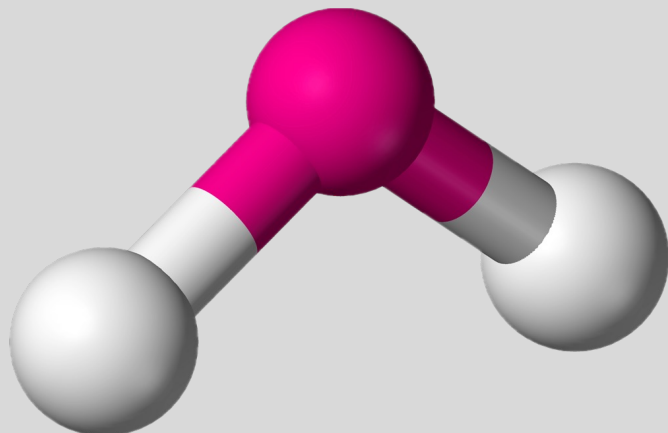
Write Lewis structure:



4 electron densities to arrange in space: tetrahedral arrangement



Final structure: **bent**

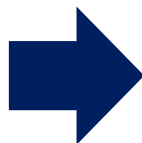
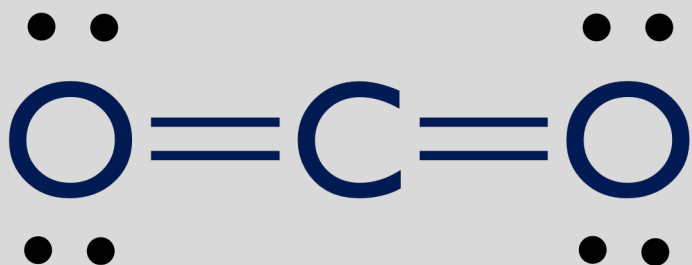


H—O—H angle *ca.* 105 deg.

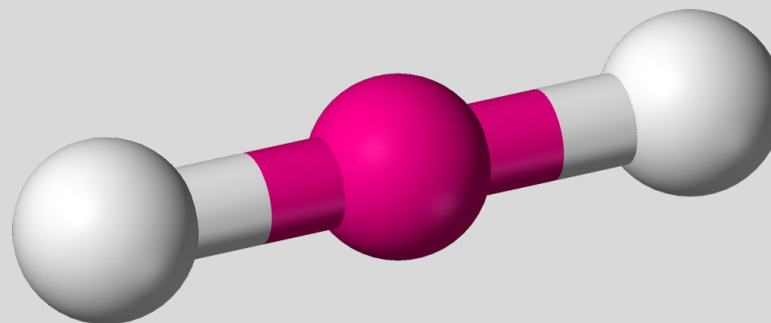
Show that water is a polar molecule!

VSEPR Example II: CO₂

Write Lewis structure:



2 electron densities to arrange in space: tetrahedral arrangement

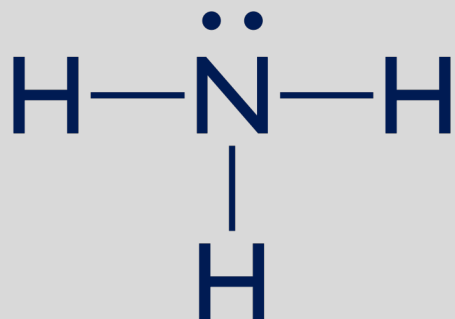


Final structure: **linear**

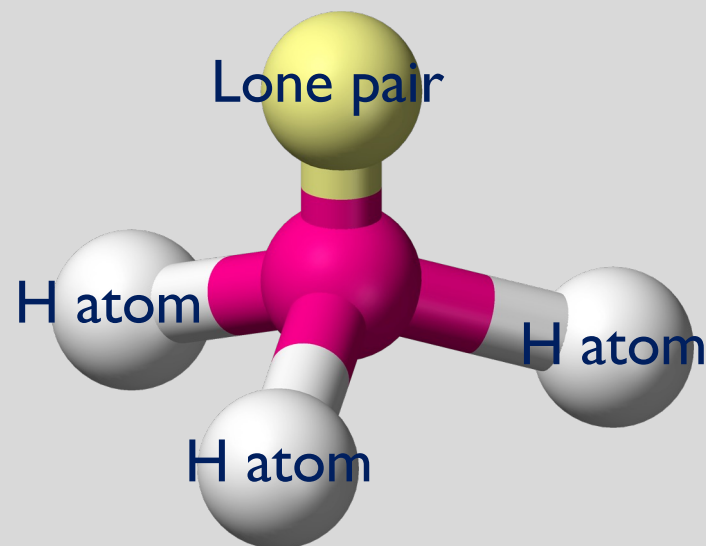
Is CO₂ a polar molecule?

VSEPR Example III: NH_3

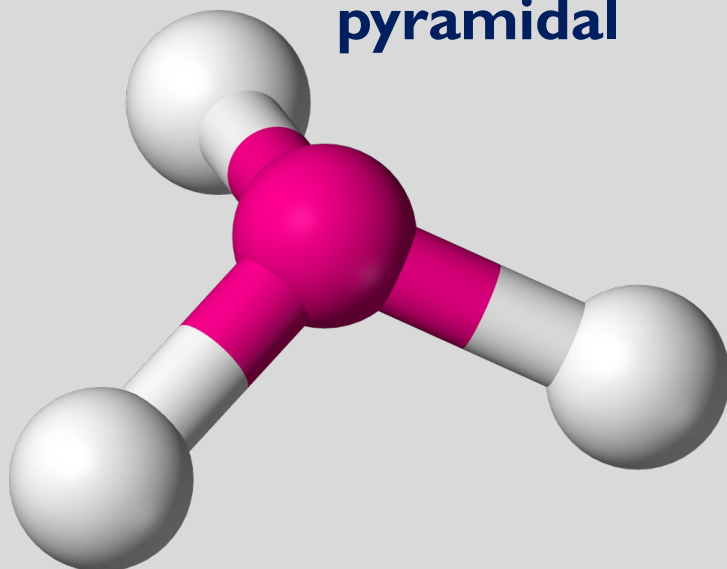
Write Lewis structure:



4 electron densities to arrange in space: tetrahedral arrangement

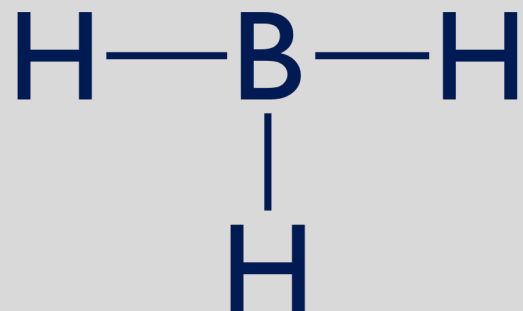


Final structure: **trigonal pyramidal**

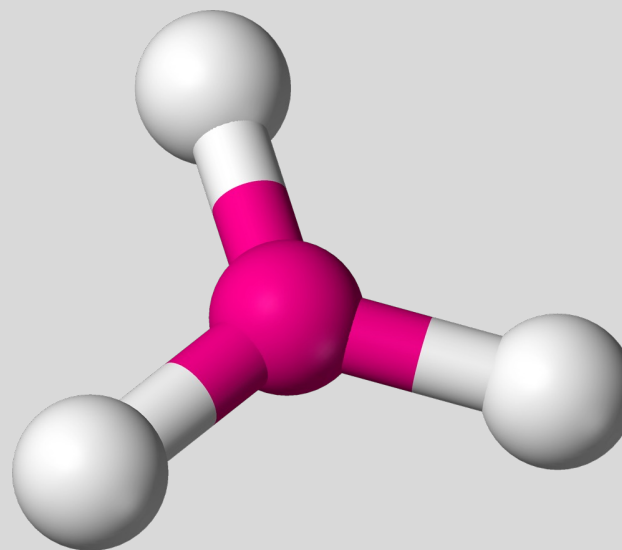


VSEPR Example IV: BH_3

Write Lewis structure:



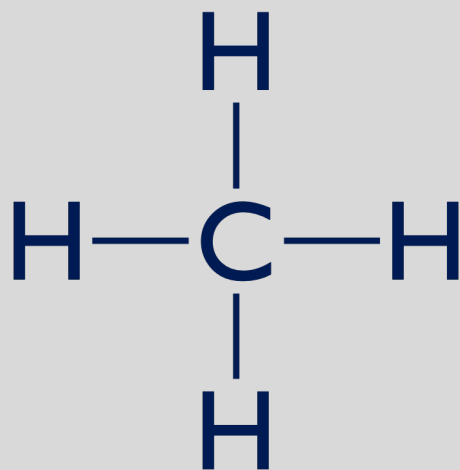
3 electron densities to arrange in space: trigonal arrangement



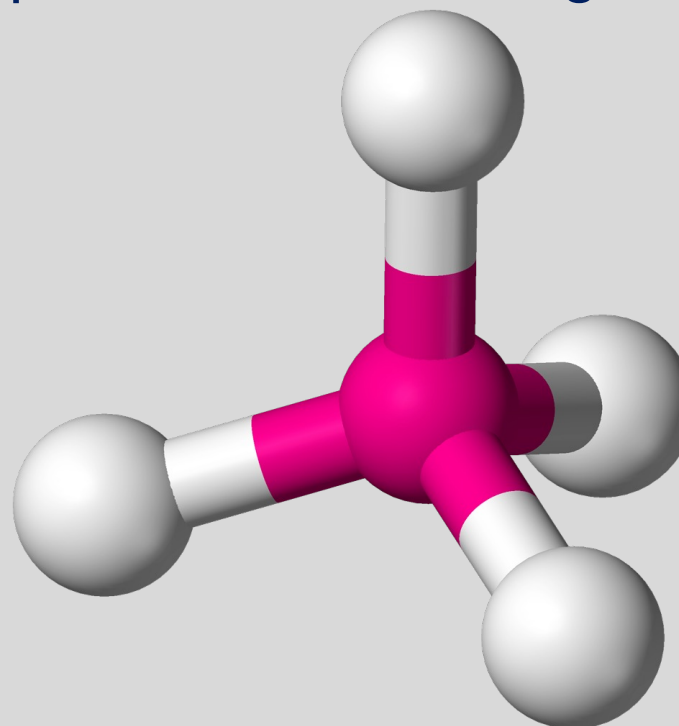
Final structure: **trigonal planar**

VSEPR Example V: CH₄

Write Lewis structure:

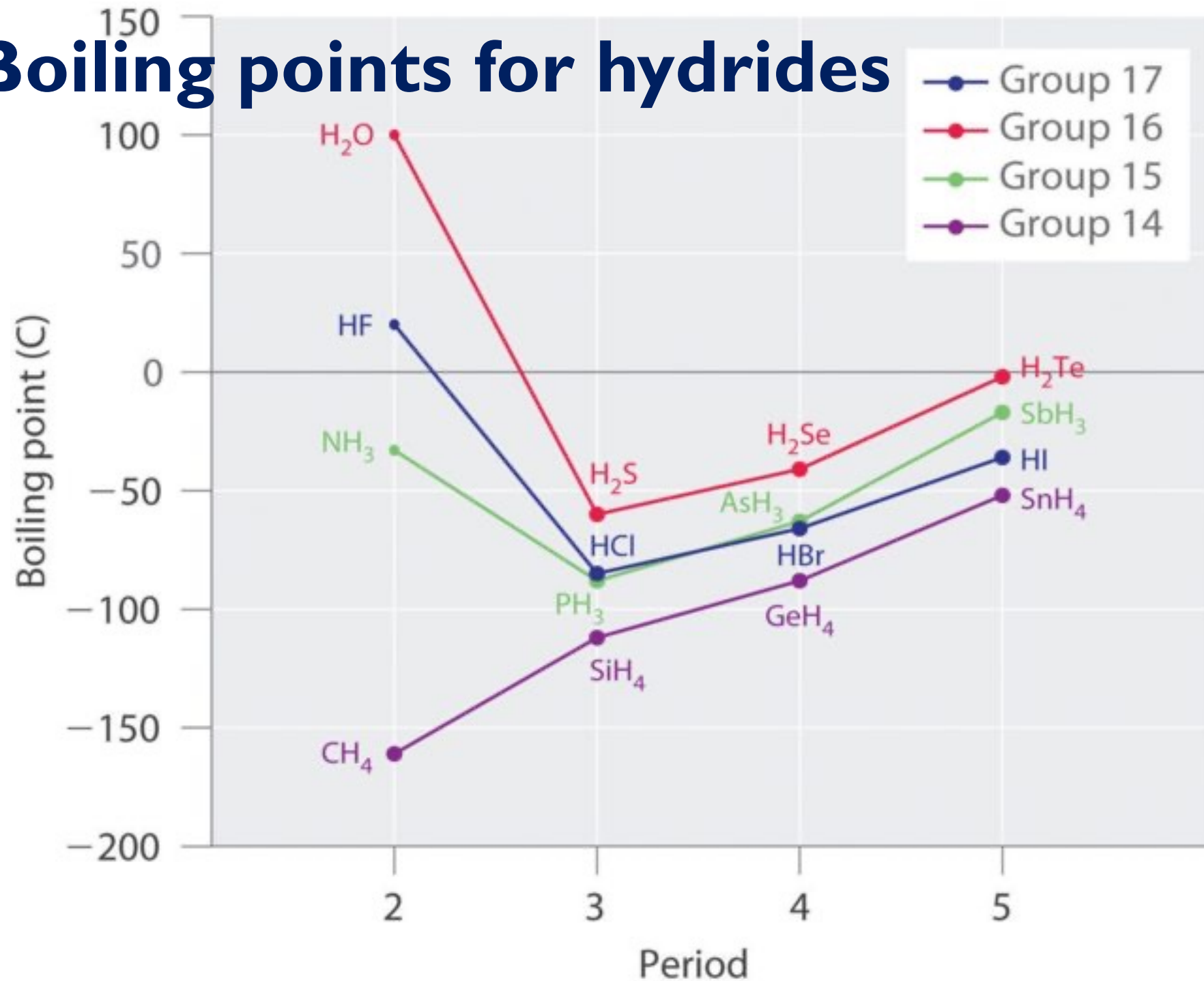


4 electron densities to arrange in space: tetrahedral arrangement



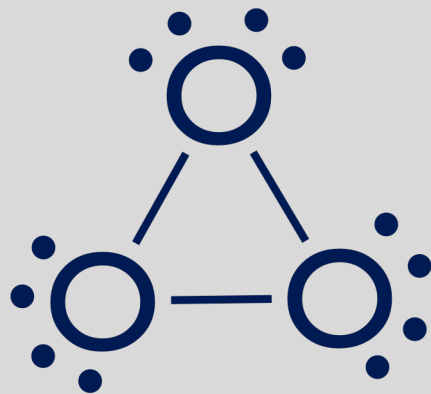
Final structure: **tetrahedral**

Boiling points for hydrides



Inadequacy of Lewis structures: Ozone, O_3

Write Lewis structure:

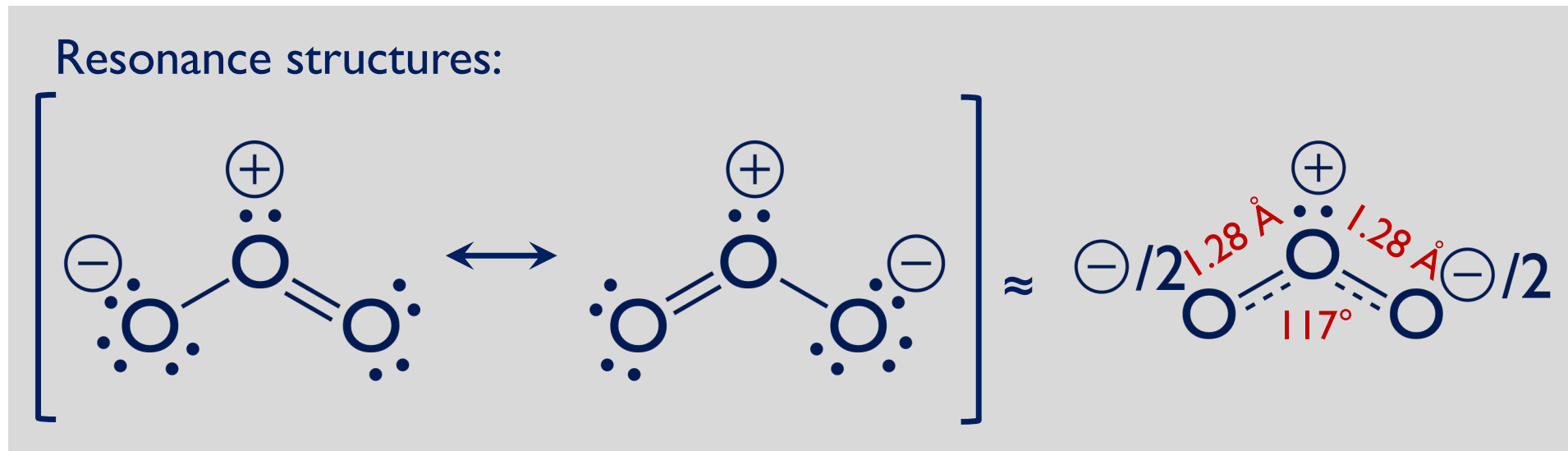


Problem: the experimentally determine O—O—O bond angle is **117°**, very far from the expected **60°**!

Still, all O—O bond lengths are identical (1.28 Å)

Inadequacy of Lewis structures:

Ozone

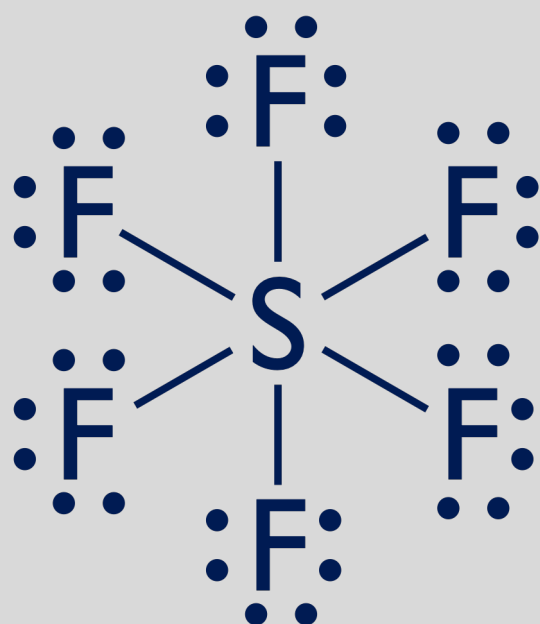


A resonance *hybrid*: the electronic ground state best described as several Lewis structures termed *resonance forms*. Real electronic structure - a mixture of the resonance forms.

Often the resonance form that carries the fewest charges and where any negative charges are localized on the more electronegative elements is favored.

VSEPR Example VI: SF_6 and the expanded octet

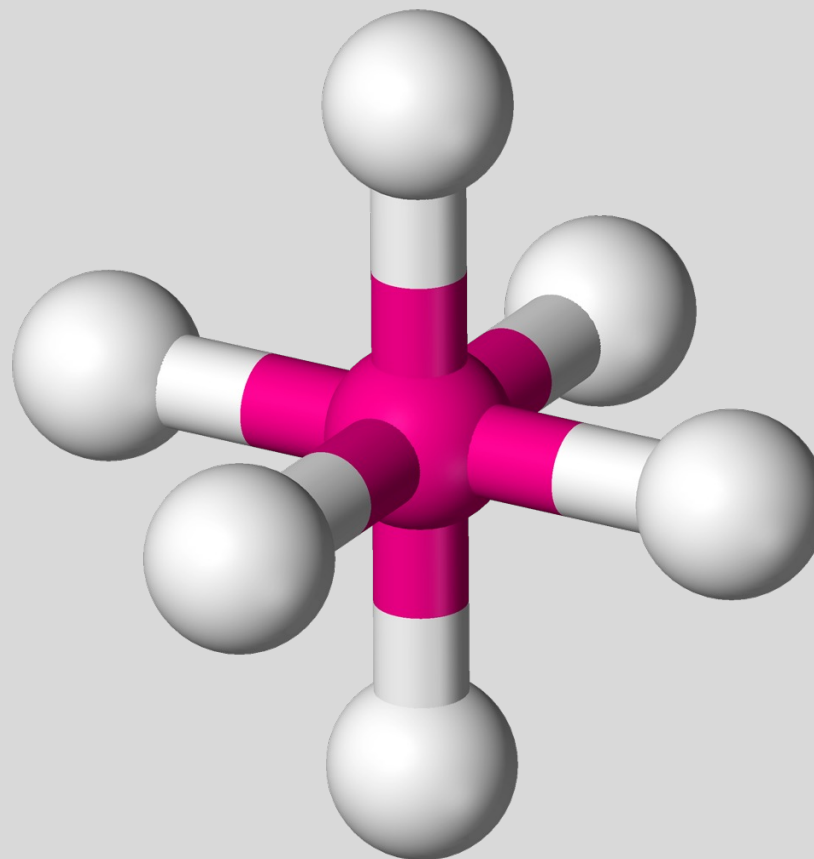
Write Lewis structure:



Sulfur surrounded by 12 e^- !
OK – not 2nd period



6 electron densities to arrange in space: octahedral arrangement



More exercises

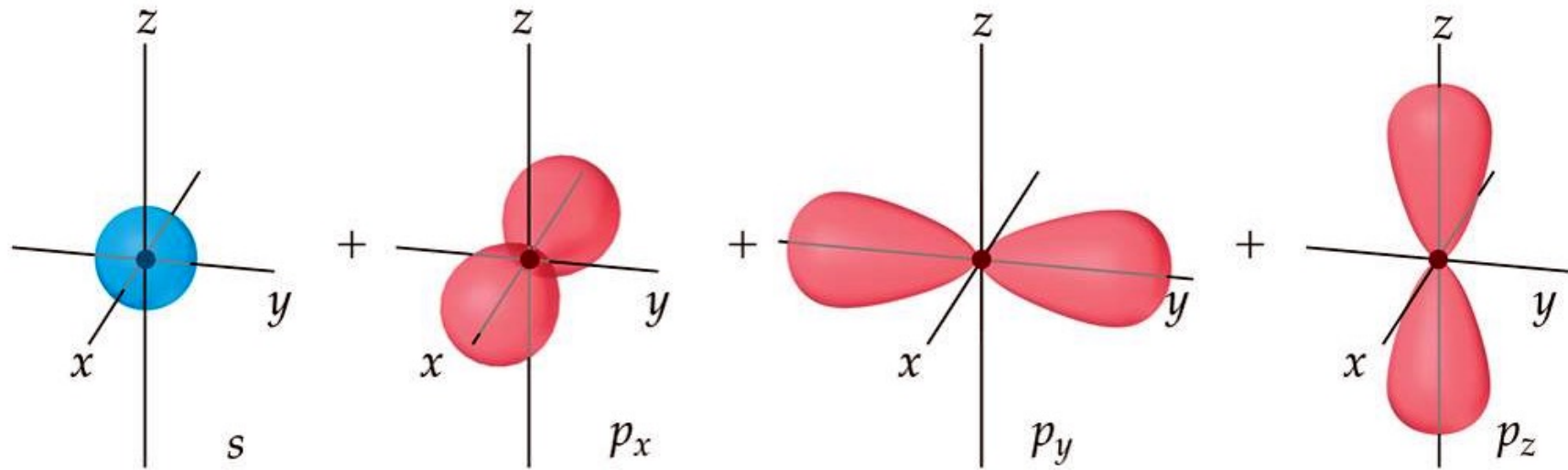
Write Lewis structures (incl. formal charges) and predict structures using the VSEPR model for:

- a) IF_5
- b) SOCl_2
- c) ClO_3^-
- d) SF_4
- e) CH_3OH
- f) XeF_2
- g) NO_2^-

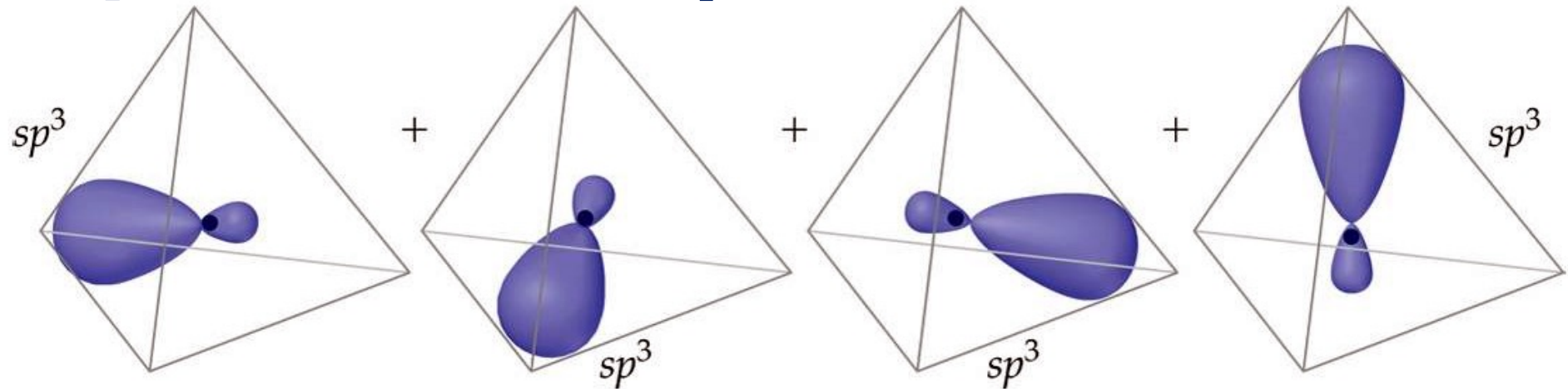
Strategy:

- 1) Write skeletal structure of molecule
- 2) Count the total number of valence electrons
- 3) Draw all bonds as single bonds and fill remaining electrons to complete octets
- 4) If it does not add up, use lone-pairs to generate double or triple bonds.

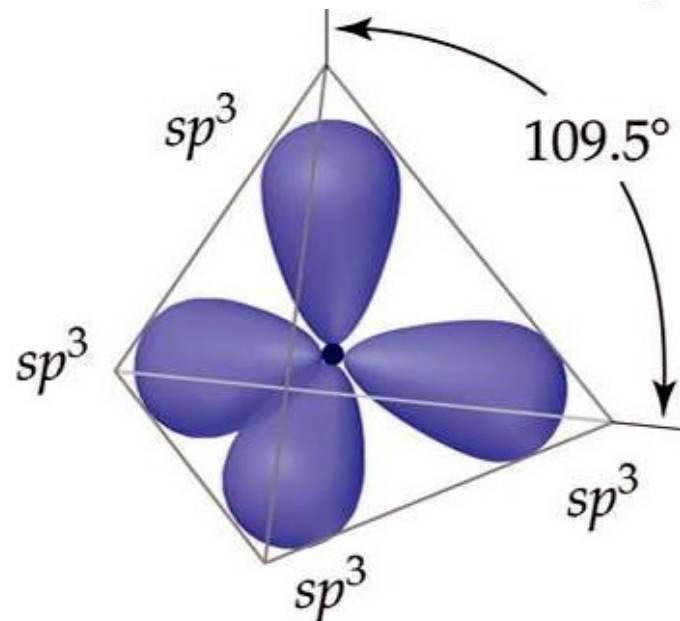
Hybridization, sp^3



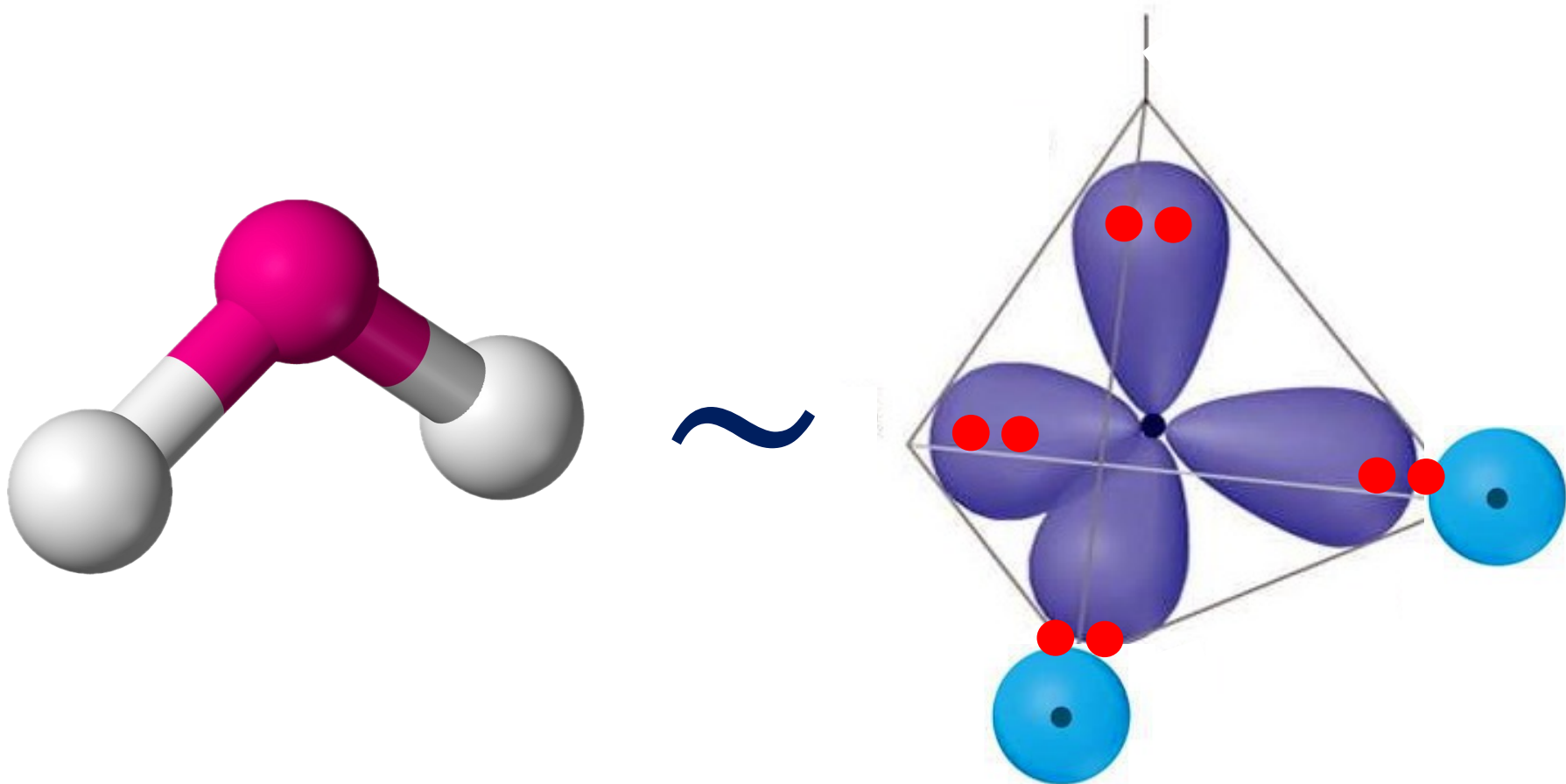
Hybridization, sp^3



Shown together (large lobes only)



Orbital overlaps

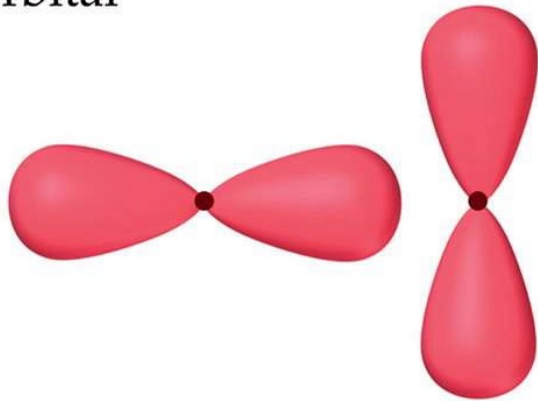


Hybridization, sp^2

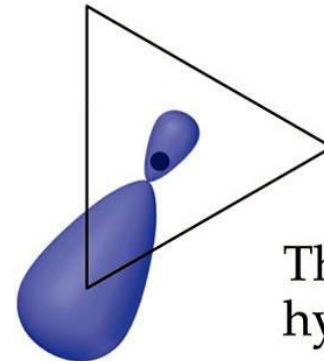
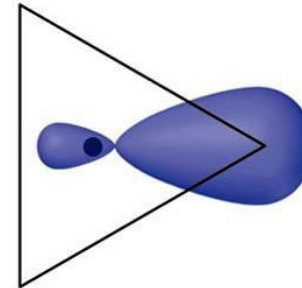
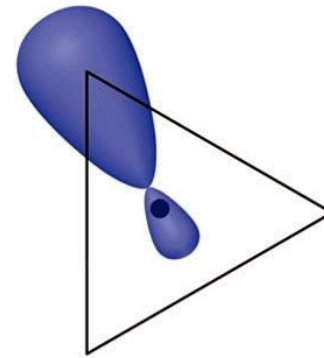
One s orbital



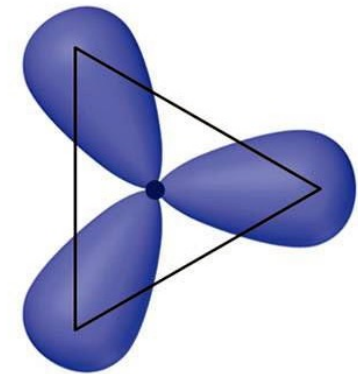
Two p orbitals



Hybridize



sp^2 hybrid orbitals
shown together
(large lobes only)



Three sp^2
hybrid orbitals

Hybridization, sp^1

